

Daily Briefing - November 14, 2026

Your Daily Tech & Programming Digest

Saturday, November 14, 2026

1000	122196	1190	35
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Easily download files from the Open Science Framework with Papercheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-22 📖 3 min 📝 765 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

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http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11
21 min
4292 words

TWENTY PERCENT STATISTICIAN

Summary: Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Astragalin ameliorates cognitive dysfunction in Alzheimer's disease model mice: Role of the iron metabolism-autophagy axis

1 min
33 words

BRAIN RESEARCH

Summary: Publication date: 1 February 2027 Source: Brain Research, Volume 1896 Author(s): Can Wang, Wanlan Huang, Weishan Zhang, Li Zhang, Chang Zhou, Yaqi Yang, Shuhan Wang, Jiahong Lin, Jing Liu, Guoying Li, Yonggang Yang, Yuxin Ma

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https://www.sciencedirect.com/science/article/pii/S0006899326004889?dgcid=rss_sd_all

Establishment of an animal model of GPIAS-defined tinnitus-like behavior with normal auditory brainstem response threshold profiles

📖 1 min 📝 24 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Rong Wang, Yangyang Yu, Qianru Yu, Rui Guo, Menghua Li, Yang Li, Shusheng Gong, Ke Liu

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https://www.sciencedirect.com/science/article/pii/S0306452226007190?dgcid=rss_sd_all

Corrigendum to “Neuroprotective effects of lixisenatide and liraglutide in the MPTP mouse model of Parkinson’s disease” [Neuroscience 303 (2015) 42–50]

📖 1 min 📝 20 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): W. Liu, J. Jalewa, M. Sharma, G. Li, L. Li, C. Hölscher

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https://www.sciencedirect.com/science/article/pii/S0306452226007220?dgcid=rss_sd_all

Desync and connect: Noninvasive MEG evidence of long-range coupling between locally desynchronized regions

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Daria Kleeva, Alexei Ossadtchi

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https://www.sciencedirect.com/science/article/pii/S1053811926006300?dgcid=rss_sd_all

Sex-related multivariate organization of glymphatic MRI markers in healthy adults

📖 1 min 📝 22 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Rui Wang, Shulan Wu, Jiajun Xu, Xiaoqi Huang, Simin Zhang, Qiyong Gong, Huaiqiang Sun

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https://www.sciencedirect.com/science/article/pii/S1053811926006348?dgcid=rss_sd_all

Astrocytic ADCY8-VEGFA signaling is required for dendritic morphogenesis, neuronal activity and contextual fear memory

📅 2026-11-14 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41418-026-01899-z

Patient-level and duplicate-image data leakage inflate reported accuracy in OASIS and KAAD MRI benchmarks

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https://www.nature.com/articles/s41598-026-77108-1

Rectal compliance and nitrergic and purinergic inhibitory neurotransmission in rectal smooth muscle in mouse

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https://www.nature.com/articles/s41598-026-77619-x

Volitional stopping is preceded by a transient motor cortex EEG beta oscillation

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https://www.nature.com/articles/s42003-026-11331-4

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

👤 Kay A. Robbins 📅 2026-10-12 📖 1 min 📝 0 words FRONTIERS NEUROINFORMATICS

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https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

👤 Han Jong Jun 📅 2026-10-20 📖 1 min 📝 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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Virtual brain models: a roadmap towards effective brain digital twins

👤 Viktor Jirsa 📅 2026-10-29 📖 1 min 📝 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

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Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

Lijuan Chen 2026-10-29 1 min 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1993109

Real-time quaking-induced conversion detection of pathological prion protein in skeletal muscle, nasal and neural tissues of cattle with natural L-BSE

Maria Mazza 2026-11-11 1 min 195 words FRONTIERS NEUROSCIENCE

Summary: IntroductionBovine spongiform encephalopathy (BSE) includes classical BSE (C-BSE) and two atypical forms, high-type (H-BSE) and low-type (L-BSE). Compared with C-BSE, the peripheral distribution and tissue tropism of atypical BSE remain poorly characterized. The apparent spontaneous occurrence of at...

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From hemodynamics to function: the role of fNIRS and hybrid EEG-fNIRS systems in next-generation assistive technologies for rehabilitation

Vidhya Sampath 2026-11-11 1 min 220 words FRONTIERS NEUROSCIENCE

Summary: Assistive technologies (AT) serve as fundamental tools for rehabilitation because they help people with neurological or physical disabilities to restore their abilities, independence and their overall life satisfaction. Existing AT control methods, which include manual operation, eye tracking, voice...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1968129

Orexinergic descending modulation of pain and itch

Tomoyuki Kuwaki 2026-11-11 1 min 246 words FRONTIERS NEUROSCIENCE

Summary: Descending pain control dynamically regulates nociceptive transmission through brainstem and spinal circuits according to behavioral and physiological demands. Within this system, accumulating evidence identifies hypothalamic orexin neurons as an important component of endogenous pain control. Orexi...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2019761

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Automated detection of blink reflexes evoked by optogenetic stimulation of TRPV1-lineage corneal afferents in transgenic mice

👤 Carl Y Saab 📅 2026-11-12 📖 1 min 📝 28 words LOW VISION

Summary: CONCLUSIONS: This video-based method enables accurate, high-throughput, and observer-independent empirical assessment of stimulus-evoked blink reflexes, providing a quantitative platform for investigating corneal sensory responses relevant to ocular nociception.

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Step change in glaucoma polygenic risk score performance enables clinical utility

👤 Jamie E Craig 📅 2026-11-12 📖 1 min 📝 58 words LOW VISION

Summary: CONCLUSIONS: This enhanced PRS, available for clinical use through an accredited genotyping workflow, enables effective risk stratification for OAG across diverse ancestries, facilitating earlier identification and treatment of high-risk individuals while supporting resource rationalisation for thos...

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A robust framework for cross-level spatial-frequency collaborative recognition in plant disease images

Haiming Lan 2026-11-12 1 min 61 words LOW VISION

Summary: Plant disease images commonly suffer from pixel confusion, texture blurring and semantic ambiguity caused by cluttered backgrounds, noise interference and diverse lesion morphologies. These intra- and inter-class disturbances break the representation balance between global context and local details,...

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Creating perceptually uniform circular shape spaces: An automated method for asymmetric naturalistic shapes

Yaniv Morgenstern 2026-11-12 1 min 63 words LOW VISION

Summary: Shape processing is crucial for higher-level visual functions such as object recognition. Yet, isolating high-level vision from low-level shape influences remains challenging. Typically, studies use simple, symmetric stimuli and labor-intensive psychophysical measurements to control for perceptual s...

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Genetic parameters and genomic prediction accuracy for piling behaviour using individual-level computer-vision data from ArUco-marked laying hens

👤 P Bijma 📅 2026-11-12 📖 1 min 📝 69 words LOW VISION

Summary: CONCLUSIONS: This study provides the first evidence that piling behaviour in laying hens is a heritable trait. The significant genetic variance and predicted response to selection across traits confirm the potential for piling mitigation through genetic selection. A practical strategy for breeding p...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

Joseph Paul Nemargut

2026-11-12 1 min 69 words LOW VISION

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Computer vision in precision poultry farming: evaluation of YOLO and transformer-based object detectors for dead broiler detection in farm inspection images

👤 Petr Bartoš 📅 2026-11-13 📖 1 min 📝 71 words LOW VISION

Summary: Broiler mortality inspection is an important component of poultry farm management. This study presents a pilot technical feasibility evaluation of recent YOLO and transformer-based object detection models for detecting dead broilers in event-enriched farm inspection images. The dataset was collected...

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Uptake of and adherence to diabetic retinopathy screening in an opportunistic setting: a quasi-experimental cohort study from Upper Austria

👤 Josef Huemer 📅 2026-11-13 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSION: Although DMPs based on an opportunistic model improve overall adherence to screening guidelines, they do not achieve the same level of participation in retinopathy screening as systematic programmes in other countries.

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Hessian-sparsity untrained neural network for near-infrared single-photon, single-pixel imaging

Fan Wang 2026-11-13 1 min 62 words LOW VISION

Summary: Near-infrared (NIR) imaging has emerged as an important technology for night vision, remote sensing, and biological imaging, yet conventional array-detector-based systems are often limited by insufficient sensitivity, high cost, and substantial dark noise. Single-pixel imaging (SPI) offers an attrac...

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Image Quality Assessment for Machine-oriented Communication

Wenjun Zhang 2026-11-13 1 min 71 words LOW VISION

Summary: The rapid development of Multimoal Large Language Models (MLLMs) has driven the evolution of communication paradigms, making the transmission of visual information in extremely low-bandwidth possible. However, as it is oriented toward computer vision tasks rather than humans, there is no widely acce...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

👤 Liu Yang 📅 2026-08-19 📖 1 min 📝 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

👤 Hideaki Onishi 📅 2026-09-25 📖 1 min 📝 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

👤 Joseph Paul Nemargut 📅 2026-11-12 📖 1 min 📝 69 words BRAILLE

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

👤 Sevda Uzun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Prefrontal transcranial direct current stimulation combined with cognitive training reduces anxiety, daytime sleepiness, and cognitive complaints in women after COVID-19: A randomized, double-blind, sham-controlled trial

👤 Guilherme Augusto Santos Bueno
📅 2026-11-12 📖 1 min 📝 37 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Adding left-prefrontal tDCS to cognitive training reduced anxiety and daytime sleepiness and improved cognition and psychomotor speed in women with post-COVID cognitive complaints, with parallel changes in prefrontal EEG markers. Confirmation in larger samples is needed.

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Home-based transcranial direct current stimulation and exercise for pain in people with knee osteoarthritis: A mixed-methods pilot randomised controlled trial

👤 Rocco Cavaleri 📅 2026-11-12 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: This mixed-methods pilot trial evaluated the feasibility and patient perceptions of home-based transcranial direct current stimulation (tDCS) combined with exercise for pain in people with knee osteoarthritis. Twenty-four participants were allocated to active or sham home-based tDCS plus exercise. P...

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Effects of high-definition transcranial direct current stimulation on muscle strength, proprioception, cutaneous sensitivity and their relationship with balance in older adults with a history of falls

👤 Wei Sun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: A 4-week HD-tDCS intervention significantly improves static and dynamic balance, hip strength, and ankle inversion proprioceptive in older adults with a history of falls. Improved static balance is positively correlated with increased hip abduction strength.

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

👤 Yicheng Zhu 📅 2026-11-10 📖 1 min 📝 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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Acute aerobic exercise and orbitofrontal cortex reactivity during positive emotion processing in obesity: An exploratory fNIRS study

👤 Jun Yan 📅 2026-11-12 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSION: Obese students showed lower baseline PA and right OFC activation during active positive elaboration; post-exercise PA and right OFC activation were higher in the same context. These single-arm associations cannot be attributed to exercise; right OFC is a preliminary trial signal, dlPFC i...

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Motor imagery enhances swallowing motor cortex excitability and activates sensorimotor regions: a TMS and fNIRS study

👤 Yue Lan 📅 2026-11-12 📖 1 min 📝 51 words FNIRS

Summary: CONCLUSION: MI enhances the excitability of the swallowing motor cortex and activates the key sensorimotor cortical areas governed ME of swallowing. MI shows greater superiority over AO and may become a promising effective rehabilitation strategy for neurogenic dysphagia. Trial registration Chinese ...

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Effects of voice and gesture interaction with in-vehicle robots on driving performance, driving workload, and user experience: evidence from multimodal physiological measures across driving environments

👤 Liting Li
📅 2026-11-13 📖 1 min 📝 61 words FNIRS

Summary: This study investigates the effects of voice and gesture interaction with in-vehicle robots on driving performance, workload, and user experience using multimodal physiological measures. A 2 (driving environment: simple vs. complex) × 3 (interaction modality: none, voice, gesture) within-subject sim...

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Altered task-evoked sensorimotor cortical hemodynamic responses in degenerative cervical myelopathy: an fNIRS pilot study

👤 Zachary A Smith
📅 2026-11-13 📖 1 min 📝 37 words FNIRS

Summary: CONCLUSIONS AND FUTURE WORK: DCM was associated with task dependent attenuation of sensorimotor cortical hemodynamic responses and impaired hand dexterity. These preliminary findings suggest altered task-evoked sensorimotor cortical hemodynamics in DCM and warrant confirmation in larger studies.

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The spatial updating mechanism of different field cognitive styles in various perspective orientations: Evidence from behavior and fNIRS

Yixue Dong 2026-11-13 1 min 65 words FNIRS

Summary: Spatial updating, the dynamic maintenance of self-environment spatial relations, is a core process in human navigation. It nevertheless remains unclear how field cognitive style (field-dependent [FD] vs. field-independent [FI]) and perspective orientation jointly shape this process. We therefore com...

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Effects of acupuncture on cognitive function and brain activation in older adults with mild cognitive impairment: A randomized controlled trial

Wenguang Xia 2026-11-13 1 min 25 words FNIRS

Summary: CONCLUSION: Acupuncture may improve cognitive function in older adults with MCI, accompanied by an increase in task-related β-values in the temporal cortex and Broca's area.

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Low-amplitude intracortical microstimulation produces short-latency modulation throughout the cortical reach-to-grasp network

Marc H Schieber 2026-11-12 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Intracortical microstimulation (ICMS) can be used to deliver information to the cortex from a bidirectional brain-machine interface (BMI), thereby improving control of neuroprosthetic devices. Primary somatosensory cortex (S1) is a common choice for delivering information, but S1-ICMS can...

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Hierarchical causal construction for semantic-granularity-aligned counterfactual medical image generation

Yifan Chen 2026-11-12 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Counterfactual image generation seeks to answer how a medical image should change when a clinical semantic factor is altered, and is important for model interpretation, lesion localization, and controllable image editing. This work addresses the semantic granularity mismatch in existing counterfactu...

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Sub-second dopamine reinforcement shapes juvenile social play

Ying Li 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Juvenile social play is critical for neurodevelopment, yet the neural mechanisms organizing moment-to-moment interactions remain elusive. Using social behavior sequencing (Social-seq)-an occlusion-robust 3D pose estimation and unsupervised clustering pipeline-we found that nucleus accumbens (NAc) do...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-11-13 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Motor imagery (MI) can support self-initiated control, whereas brain-computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspend...

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Optogenetics in neurobiology and neurodegenerative disorders: from neural circuit mechanisms to precision neuromodulation

👤 V Singaravel 📅 2026-11-13 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: Neurodegenerative disorders, including Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis, multiple sclerosis and stroke, are characterized by progressive neuronal dysfunction, synaptic impairment and neural circuit degeneration, leading to severe cognitive...

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Brain-computer interfaces for neurodegenerative diseases: A comprehensive review of closed-loop systems from diagnosis to neural rehabilitation

👤 Yadong Guo 📅 2026-11-13 📖 1 min 📝 70 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) technology serves as an important bridge between the nervous system and external devices and has shown considerable promise for the diagnosis and treatment of neurodegenerative diseases, whose global burden continues to increase: in 2021, more than 57 million people we...

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Psychological Predictors of Pilot Performance: An Integrative Review of Proximal and Distal Performance Criteria

👤 Xiang Yao 📅 2026-11-13 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Research on pilots' psychological attributes and performance spans diverse predictors and performance criteria, but the findings remain fragmented. This integrative review synthesizes 60 empirical studies using a predictor-criterion framework that distinguishes cognitive abilities and behavioral dis...

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Cortical Network Alterations and Treatment-Induced Reorganization in Panic Disorder: A Longitudinal High-Density EEG Study

Yuan Yang 2026-11-13 1 min 33 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: PD is characterized by frequency-specific cortical network disruptions that show dynamic reorganization following treatment. These findings highlight the potential of EEG-derived network features as candidate markers for PD identification and treatment monitoring.

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A User-Function Taxonomy for Brain-Computer Interfaces: From Task Definition to Evidence

Xiaorong Gao 2026-11-13 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) studies are commonly compared by electrode, signal, paradigm, device, or clinical purpose. These descriptors do not identify the user-level function implemented in a task, although that function determines the appropriate endpoint. This article proposes the Type-Wield-...

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GB-algebraic class

Jiangang Tang 2026-11-13 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: The purpose of this article is to study the relationships among the BCI-algebraic class, the GB-algebraic class, and the class of algebras of type (2,0). The results indicate that: (1) The BCH-algebraic class and the GB-algebraic class are two distinct algebraic classes, both of which contain the BC...

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Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

2026-09-10 1 min 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 819 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 852 words FMHY

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https://fmhy.net/posts/may-2026

Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1025 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the polls a...

🔗 **Read full article:**
https://fmhy.net/posts/oct-2026

Monthly Updates [November]

📅 2026-11-01 📖 4 min 📝 933 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/nov-2026

whiff-python 0.1.0: one JSON report for a file or a blob, from the Rust whiff crate

👤 /u/Shoddy_One4465 📅 2026-11-14 📖 1 min 📝 210 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> whiff reads a file and returns one JSON document: format, hashes, rights, risk signals (macros, phishing, embedded secrets), and a compact content view. The Rust crate covers the default format collection in-house, with budgets on reads, decompression, and time, and...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xvv4z8/whiffpython_010_one_json_report_for_a_file_or_a/

Show HN: Live map of U.S. flights during tonight's FAA nationwide ground stop

👤 gvsi 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://grounds.top/ Comments URL: https://news.ycombinator.com/item?id=50491545 Points: 133 # Comments: 18

🔗 **Read full article:**
https://grounds.top/

Unsloth Qwen3.8-27B Quantizations – Dynamic v3.1 Update

👤 Retro_Dev 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://unsloth.ai/docs/models/qwen3.8 Comments URL: https://news.ycombinator.com/item?id=50491980 Points: 113 # Comments: 25

🔗 **Read full article:**
https://unsloth.ai/docs/models/qwen3.8

Philip Roth and David Remnick getting personal (2001)

👤 Caiero 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://salmagundi.skidmore.edu/articles/2001-philip-roth-david-remnick Comments URL: https://news.ycombinator.com/item?id=50492098...

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https://salmagundi.skidmore.edu/articles/2001-philip-roth-david-remnick

Elephants found to be increasingly isolated leading to fractured social networks

👤 gmays 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://today.ucsd.edu/story/elephants-found-to-be-increasingly-isolated-leading-to-fractured-social-networks Comments URL: <a href="https://news.ycombin...

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https://today.ucsd.edu/story/elephants-found-to-be-increasingly-isolated-leading-to-fractured-social-networks

Security Agencies Worldwide Found to Use Ad-Based Surveillance System

👤 billybuckwheat 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://citizenlab.ca/security-agencies-worldwide-found-to-use-ad-based-surveillance-system/ Comments URL: <a href="https://news.ycombinator.com/item?id=50492950">https:/...

🔗 **Read full article:**
https://citizenlab.ca/security-agencies-worldwide-found-to-use-ad-based-surveillance-system/

Ggml v1.0.0

👤 cobychapple 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/ggml-org/ggml/releases/tag/v1.0.0 Comments URL: https://news.ycombinator.com/item?id=50493062 Points: 222 # Comments:...

🔗 **Read full article:**
https://github.com/ggml-org/ggml/releases/tag/v1.0.0

Nvidia allegedly engineered scarcity by funding AI startups to buy its GPUs

👤 speckx 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.techpowerup.com/354814/nvidia-allegedly-engineered-scarcity-by-funding-ai-startups-to-buy-its-gpus Comments URL: <a href="https://news.ycomb...

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https://www.techpowerup.com/354814/nvidia-allegedly-engineered-scarcity-by-funding-ai-startups-to-buy-its-gpus

Monkey Island: all 13 versions side by side

👤 maximilianburke 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://monkeyisland.pages.dev/ Comments URL: https://news.ycombinator.com/item?id=50493904 Points: 40 # Comments: 4

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https://monkeyisland.pages.dev/

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

Adriel Carridice 2025-11-04 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Paclitaxel induces region-specific microglial remodeling associated with chronic behavioral deficits and modulated by electroacupuncture in rats

1 min 19 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Shinya Hiraiwa, Taiga Muramoto, Masahiro Umeda, Satomi Ebara, Fumihiko Fukuda

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004828?dgcid=rss_sd_all

Gastrin-releasing peptide in the anterior medial nucleus accumbens or paraventricular nucleus suppresses chow intake under food restriction but not palatable diet intake in nonrestricted rats

📖 1 min 📝 28 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Regan Cochran, Ella Irvin, Yiwei Tang, Stacey Chuang, Adele Nadeau, Smruti Patel, Lindsey Wilson, Weijia Zhang, Wayne E. Pratt

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004877?dgcid=rss_sd_all

SREBP1 inhibition downregulates NR2B-PSD95 interaction in glutamate excitotoxicity following cerebral ischemic reperfusion injury

📖 1 min 📝 16 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Lopmudra P. Sarode, Bhavesh Verma, Rajesh Ugale

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004841?dgcid=rss_sd_all

Learning uncertainty fields for joint brain stroke diagnosis and intrinsic lesion localization in CT and MRI

📅 2026-11-14 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41746-026-03458-9

Low-frequency phase in early visual cortex gates prefrontal influence on perception

📅 2026-11-14 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11359-6

Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

👤 Roman J. Giger
📅 2026-11-05 📖 1 min 📝 21 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 05 November 2026; doi:10.1038/s41593-026-02534-3 Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02534-3

The purinergic receptor P2Y13 controls the activation and mode of division of adult ventricular–subventricular zone neural stem cells

👤 Lucía Paniagua-HerranzJulia Serrano-LópezCelia Llorente-SáezMarina Leonor Pérez-SanzDavid de Agustín-DuránPere Duart-AbadiaAna Domingo-MuelasÁfrica Vincelle-NietoPaloma BragadoDiana Manzano-FrancoMarina Arribas-BlázquezLucía GallegoAraceli Garcia-MoraLuis Alcides Olivos-OréSergio GascónÁlvaro Gutierrez-UzquizaBenedikt BerningerRaquel Pérez-SenEsmerilda G. DelicadoRosa Gómez-VillafuertesArmando Reyes-PalomaresAntonio R. ArtalejoIsabel FariñasFelipe Ortegaahttps://ror.org/02p0gd045Departamento de Bioquímica y Biología Molecular, Universidad Complutense de Madrid, Madrid 28040, SpainbInstituto Universitario de Investigación en Neuroquímica, Departamento de Bioquímica y Biología Molecular, Madrid 28040, Spainchttps://ror.org/014v12a39Instituto de Investigación Sanitaria del Hospital Clínico San Carlos de Madrid, Madrid 28040, Spaindhttps://ror.org/02p0gd045Departamento de Farmacología y Toxicología, Facultad de Veterinaria, Universidad Complutense de Madrid, Madrid 28040, Spainehttps://ror.org/043nxc105Departamento de Biología Celular y Biología Funcional, Universidad de Valencia, Valencia 46100, Spainfhttps://ror.org/043nxc105Instituto de Biotecnología y Biomedicina, Departamento de Genética, Facultad de Biología, Universidad de Valencia, Valencia 46100, Spainghttps://ror.org/02g87qh62Centro de Investigación Biomédica en Red sobre Enfermedades Neurodegenerativas, Departamento de Biología Celular y Biología Funcional, Valencia 46100, Spainhhttps://ror.org/02gfc7t72Cajal Neuroscience Center, Laboratory of direct reprogramming, Spanish National Research Council, Madrid 28805, Spainihttps://ror.org/0220mzb33Centre for Developmental Neurobiology, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomjhttps://ror.org/0220mzb33Medical Research Council Centre for Neurodevelopmental Disorders, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomkhttps://ror.org/023b0x485Institute of Physiological Chemistry, University Medical Center Johannes Gutenberg University, Mainz 55128, GermanylFocus Program Translational Neuroscience, Johannes Gutenberg University, 55128, Mainz, Germanymhttps://ror.org/036b2ww28Department of Molecular Biology and Biochemistry, Faculty of Science, University of Malaga, Malaga SE1 1UL, Spain

📅 2026-11-04 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceAdult neural stem cells (NSCs) sustain lifelong neurogenesis in the mammalian brain, yet the mechanisms controlling their activation and self-renewal remain poorly understood. We identify the puri...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2618940123?af=R

Dissociation of gravity and limb dynamics in motor control

Valeriya GritsenkoTjaša KunavarMarko JamšekSergiy YakovenkoOlivier WhiteJan Babičahttps://ror.org/011vxgd24Departments of Human Performance/Neuroscience, West Virginia University, Rockefeller Neuroscience Institute, Morgantown, WV 26506bLaboratory for Neuromechanics and Biorobotics, Department of Automatics, Biocybernetics, and Robotics, Jožef Stefan Institute, Ljubljana 1000, Sloveniachttps://ror.org/02vjkv261Université Bourgogne Europe, Institut national de la santé et de la recherche médicale, Cognition, Action and Sensorimotor Plasticity, Unité Mixte de Recherche 1093, Dijon 21000, France

📅 2026-11-03 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceThe nervous system must control movement in presence of variable intrinsic and extrinsic forces, yet it has remained unclear whether mechanically equivalent forces arising from gravity and limb dy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2613283123?af=R

Lead impairs long-term memory by disrupting mitochondrial redox homeostasis in Drosophila

Cheng-Tzu HsuTe-Wei LuYa-Ting YangChun-Chao ChenYen-Hua ChiuJung-Hsuan YangZheng-Wen WuWei-Kai ChengJheng-Wei WuChen-Hao ShiuSih-Kai ChuangKuan-Lin FengHsuan-Wen LinJia-Ling YangAnn-Shyn Chiangahttps://ror.org/00zdnkx70Institute of Biotechnology, College of Life Sciences and Medicine, National Tsing-Hua University, Hsinchu 30013, Taiwanbhttps://ror.org/00mjawt10Institute of Precision Medicine, College of Medicine, National Sun Yat-sen University, Kaohsiung 804201, Taiwanchttps://ror.org/00mjawt10Department of Biological Sciences, National Sun Yat-sen University, Kaohsiung 804201, Taiwandhttps://ror.org/00zdnkx70Brain Research Center, National Tsing Hua University, Hsinchu 30013, Taiwanehttps://ror.org/00mjawt10Department of Biomedical Science and Technology, College of Medicine, National Sun Yat-sen University, Kaohsiung 804201, Taiwanfhttps://ror.org/00zdnkx70Institute of Systems Neuroscience, College of Life Sciences and Medicine, National Tsing Hua University, Hsinchu 30013, TaiwangGraduate Institute of Biomedical Sciences, China Medical University, Taichung 40402, TaiwanhInstitute of Molecular and Genomic Medicine, National Health Research Institutes, Miaoli 35053, Taiwanihttps://ror.org/03gk81f96Department of Biomedical Science and Environmental Biology, Kaohsiung Medical University, Kaohsiung 80780, Taiwan

📅 2026-11-04 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceEnvironmental lead exposure is strongly associated with cognitive deficits, yet how it disrupts memory at the cellular level remains unclear. Here, we show that lead selectively impairs protein-sy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2612578123?af=R

Hypothalamic lncRNA Synage prevents age-related sperm quality decline by suppressing A-type potassium currents in GnRH neurons

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📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceMen increasingly delay fatherhood, yet paternal aging lowers sperm quality and increases health risks to offspring. The mechanisms underlying these reproductive changes remain incompletely defined...

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https://www.pnas.org/doi/abs/10.1073/pnas.2612408123?af=R

Transcranial magnetic stimulation to the dorsolateral prefrontal cortex modulates single-neuron activity in humans

👤 Charles W. DickeyUmair HassanHiroto KawasakiAriane E. RhoneChristopher C. ClineMatthew A. HowardNicholas T. TrappAaron D. BoesJoel I. BergerCorey J. Kellerahttps://ror.org/00f54p054Department of Psychiatry and Behavioral Sciences, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Wu Tsai Neurosciences Institute, Stanford University, Stanford, CA 94305chttps://ror.org/036jqmy94Department of Neurosurgery, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Department of Psychiatry, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Department of Neurology, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Department of Pediatrics, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Iowa Neuroscience Institute, Carver College of Medicine, University of Iowa, Iowa City, IA 52242hVeterans Affairs Palo Alto Healthcare System, and the Sierra Pacific Mental Illness Research, Education, and Clinic Center, Palo Alto, CA 94305

📅 2026-11-02 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceTranscranial magnetic stimulation (TMS) to the dorsolateral prefrontal cortex is a common treatment for depression. However, it remains unknown how TMS affects single neurons in deep brain circuit...

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https://www.pnas.org/doi/abs/10.1073/pnas.2610803123?af=R

Hepatocyte-derived exosomal miR-122 mediates liver-to-brain crosstalk and neuroinflammation in alcohol-exposed Alzheimer’s disease mice

👤 Veronika BrezaniMarti Ortega-RiberaNeda RezagholizadehRadhika JoshiViliam BrezaniMrigya BabutaBruna de Araujo Cardoso dos SantosAdam ZivnyPrashanth Thevkar NageshIonita GhiranElizabeth Michelle ShreyerScott Brian MinchenbergAllyson FlavinGyongyi Szaboahttps://ror.org/04drvxt59Division of Gastroenterology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215bHarvard Medical School, Boston, MA 02115chttps://ror.org/04drvxt59Department of Anesthesia, Critical Care and Pain Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215dhttps://ror.org/05a0ya142Broad Institute of MIT and Harvard, Cambridge, MA 02142

📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceExcessive alcohol use damages both the liver and the brain, but how these organs communicate during injury remains unclear. Our study identifies a previously unrecognized liver-to-brain communicat...

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Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka 2026-10-23 1 min 318 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

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Reporting framework for event-based model diagnostics

Cassie S. Mitchell 2026-11-11 1 min 211 words FRONTIERS NEUROINFORMATICS

Summary: Event-Based Models (EBMs) are widely used to infer latent disease progression from cross-sectional biomarker data, with existing studies employing diverse approaches to characterize uncertainty, stability, and validation. However, comparatively little guidance exists for systematically integrating a...

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SDCL-AD: a structure-aware dual-graph contrastive learning framework for Alzheimer's disease prediction using multi-omics data

Miaomiao Sun 2026-11-11 1 min 201 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionEarly prediction of Alzheimer's disease (AD) is of great significance for disease intervention, personalized diagnosis, and treatment. With the rapid accumulation of multi-omics data, integrating complementary information from different molecular layers has become a key challenge in impr...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1973333

An EMG-driven virtual reality platform for physiology-based motor interaction: technical and functional validation

D. Sili 2026-11-11 1 min 351 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: Harnessing experience-dependent neuroplasticity has become a central goal of modern neurorehabilitation. Immersive virtual reality (VR) has emerged as a promising tool, enabling task-oriented training in ecologically valid scenarios that are difficult to reproduce in conventional clinical settings. ...

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The role of the right temporal lobe in neural network mechanisms during working memory encoding

Yijun Song 2026-11-11 1 min 380 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundVisual working memory (vWM) impairment is common in patients with temporal lobe epilepsy (TLE) and is more severe in those with right TLE (RTLE), suggesting the significant role of right temporal lobe in WM processes.MethodsWe recruited patients with left TLE (LTLE group, n = 20), RTLE (RT...

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Frequency-embedded sequence learning for multi-class EEG-to-text interface

Alistair McEwan 2026-11-11 1 min 227 words FRONTIERS HUMAN NEUROSCIENCE

Summary: A brain-computer interface (BCI) is an emerging technology for decoding users' intentions, enabling speech decoding from electroencephalography (EEG) signals. This study proposes a multi-class (i.e., 5 words or phrases) EEG-to-Text classification pipeline based on a lightweight 1−dimensional convolu...

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Rethinking cerebral visual impairment: Unitary and Umbrella conceptualizations as a potential way forward

John Ravenscroft 2026-11-11 1 min 303 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Over recent decades, understanding and recognition of cerebral visual impairment (CVI) have increased considerably. CVI is now recognized as the likely leading cause of certifiable childhood visual impairment in both high and low-income countries. However, differences remain concerning its definitio...

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Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

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Efficacy of mindfulness-based interventions and biofeedback for migraine management: a systematic review and meta-analysis of randomized controlled trials

Irene A. Garcia-Yu
2026-11-11 1 min 292 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveThis systematic review and meta-analysis of randomized controlled trials aims to analyze the efficacy of mindfulness-based interventions and biofeedback on headache frequency, pain intensity, headache-related disability, psychological distress, and quality of life in adult patients with mig...

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Influence of out-of-plane electric fields in an ex vivo model of transcranial electrical stimulation

Yuhyun Lee, Vishal Jain, Yuxin Guo, Maysamreza Chamanzar, Pulkit Grover and Mats Forssell

2026-11-11 1 min 226 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Electrical stimulation is the most commonly used modality for neuromodulation, with a number of clinical and scientific applications. Although the basic mechanisms of neural activation by electrical stimulation are well understood, important questions remain regarding neural responses to ...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Automated detection of blink reflexes evoked by optogenetic stimulation of TRPV1-lineage corneal afferents in transgenic mice

👤 Carl Y Saab 📅 2026-11-12 📖 1 min 📝 28 words LOW VISION

Summary: CONCLUSIONS: This video-based method enables accurate, high-throughput, and observer-independent empirical assessment of stimulus-evoked blink reflexes, providing a quantitative platform for investigating corneal sensory responses relevant to ocular nociception.

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Step change in glaucoma polygenic risk score performance enables clinical utility

👤 Jamie E Craig 📅 2026-11-12 📖 1 min 📝 58 words LOW VISION

Summary: CONCLUSIONS: This enhanced PRS, available for clinical use through an accredited genotyping workflow, enables effective risk stratification for OAG across diverse ancestries, facilitating earlier identification and treatment of high-risk individuals while supporting resource rationalisation for thos...

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A robust framework for cross-level spatial-frequency collaborative recognition in plant disease images

Haiming Lan 2026-11-12 1 min 61 words LOW VISION

Summary: Plant disease images commonly suffer from pixel confusion, texture blurring and semantic ambiguity caused by cluttered backgrounds, noise interference and diverse lesion morphologies. These intra- and inter-class disturbances break the representation balance between global context and local details,...

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Creating perceptually uniform circular shape spaces: An automated method for asymmetric naturalistic shapes

Yaniv Morgenstern 2026-11-12 1 min 63 words LOW VISION

Summary: Shape processing is crucial for higher-level visual functions such as object recognition. Yet, isolating high-level vision from low-level shape influences remains challenging. Typically, studies use simple, symmetric stimuli and labor-intensive psychophysical measurements to control for perceptual s...

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Genetic parameters and genomic prediction accuracy for piling behaviour using individual-level computer-vision data from ArUco-marked laying hens

👤 P Bijma 📅 2026-11-12 📖 1 min 📝 69 words LOW VISION

Summary: CONCLUSIONS: This study provides the first evidence that piling behaviour in laying hens is a heritable trait. The significant genetic variance and predicted response to selection across traits confirm the potential for piling mitigation through genetic selection. A practical strategy for breeding p...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

👤 Joseph Paul Nemargut
📅 2026-11-12 📖 1 min 📝 69 words

LOW VISION

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Computer vision in precision poultry farming: evaluation of YOLO and transformer-based object detectors for dead broiler detection in farm inspection images

👤 Petr Bartoš 📅 2026-11-13 📖 1 min 📝 71 words LOW VISION

Summary: Broiler mortality inspection is an important component of poultry farm management. This study presents a pilot technical feasibility evaluation of recent YOLO and transformer-based object detection models for detecting dead broilers in event-enriched farm inspection images. The dataset was collected...

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Uptake of and adherence to diabetic retinopathy screening in an opportunistic setting: a quasi-experimental cohort study from Upper Austria

👤 Josef Huemer 📅 2026-11-13 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSION: Although DMPs based on an opportunistic model improve overall adherence to screening guidelines, they do not achieve the same level of participation in retinopathy screening as systematic programmes in other countries.

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Hessian-sparsity untrained neural network for near-infrared single-photon, single-pixel imaging

Fan Wang 2026-11-13 1 min 62 words LOW VISION

Summary: Near-infrared (NIR) imaging has emerged as an important technology for night vision, remote sensing, and biological imaging, yet conventional array-detector-based systems are often limited by insufficient sensitivity, high cost, and substantial dark noise. Single-pixel imaging (SPI) offers an attrac...

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Image Quality Assessment for Machine-oriented Communication

Wenjun Zhang 2026-11-13 1 min 71 words LOW VISION

Summary: The rapid development of Multimoal Large Language Models (MLLMs) has driven the evolution of communication paradigms, making the transmission of visual information in extremely low-bandwidth possible. However, as it is oriented toward computer vision tasks rather than humans, there is no widely acce...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

👤 Joseph Paul Nemargut 📅 2026-11-12 📖 1 min 📝 69 words BRAILLE

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

👤 Sevda Uzun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Prefrontal transcranial direct current stimulation combined with cognitive training reduces anxiety, daytime sleepiness, and cognitive complaints in women after COVID-19: A randomized, double-blind, sham-controlled trial

👤 Guilherme Augusto Santos Bueno
📅 2026-11-12 📖 1 min 📝 37 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Adding left-prefrontal tDCS to cognitive training reduced anxiety and daytime sleepiness and improved cognition and psychomotor speed in women with post-COVID cognitive complaints, with parallel changes in prefrontal EEG markers. Confirmation in larger samples is needed.

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Home-based transcranial direct current stimulation and exercise for pain in people with knee osteoarthritis: A mixed-methods pilot randomised controlled trial

👤 Rocco Cavaleri 📅 2026-11-12 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: This mixed-methods pilot trial evaluated the feasibility and patient perceptions of home-based transcranial direct current stimulation (tDCS) combined with exercise for pain in people with knee osteoarthritis. Twenty-four participants were allocated to active or sham home-based tDCS plus exercise. P...

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Effects of high-definition transcranial direct current stimulation on muscle strength, proprioception, cutaneous sensitivity and their relationship with balance in older adults with a history of falls

👤 Wei Sun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: A 4-week HD-tDCS intervention significantly improves static and dynamic balance, hip strength, and ankle inversion proprioceptive in older adults with a history of falls. Improved static balance is positively correlated with increased hip abduction strength.

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

👤 Xiaozhou Zhou
📅 2026-11-07 📖 1 min 📝 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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Acute aerobic exercise and orbitofrontal cortex reactivity during positive emotion processing in obesity: An exploratory fNIRS study

👤 Jun Yan 📅 2026-11-12 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSION: Obese students showed lower baseline PA and right OFC activation during active positive elaboration; post-exercise PA and right OFC activation were higher in the same context. These single-arm associations cannot be attributed to exercise; right OFC is a preliminary trial signal, dlPFC i...

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Motor imagery enhances swallowing motor cortex excitability and activates sensorimotor regions: a TMS and fNIRS study

👤 Yue Lan 📅 2026-11-12 📖 1 min 📝 51 words FNIRS

Summary: CONCLUSION: MI enhances the excitability of the swallowing motor cortex and activates the key sensorimotor cortical areas governed ME of swallowing. MI shows greater superiority over AO and may become a promising effective rehabilitation strategy for neurogenic dysphagia. Trial registration Chinese ...

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Effects of voice and gesture interaction with in-vehicle robots on driving performance, driving workload, and user experience: evidence from multimodal physiological measures across driving environments

👤 Liting Li
📅 2026-11-13 📖 1 min 📝 61 words FNIRS

Summary: This study investigates the effects of voice and gesture interaction with in-vehicle robots on driving performance, workload, and user experience using multimodal physiological measures. A 2 (driving environment: simple vs. complex) × 3 (interaction modality: none, voice, gesture) within-subject sim...

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Low-amplitude intracortical microstimulation produces short-latency modulation throughout the cortical reach-to-grasp network

Marc H Schieber 2026-11-12 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Intracortical microstimulation (ICMS) can be used to deliver information to the cortex from a bidirectional brain-machine interface (BMI), thereby improving control of neuroprosthetic devices. Primary somatosensory cortex (S1) is a common choice for delivering information, but S1-ICMS can...

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Hierarchical causal construction for semantic-granularity-aligned counterfactual medical image generation

Yifan Chen 2026-11-12 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Counterfactual image generation seeks to answer how a medical image should change when a clinical semantic factor is altered, and is important for model interpretation, lesion localization, and controllable image editing. This work addresses the semantic granularity mismatch in existing counterfactu...

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Sub-second dopamine reinforcement shapes juvenile social play

Ying Li 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Juvenile social play is critical for neurodevelopment, yet the neural mechanisms organizing moment-to-moment interactions remain elusive. Using social behavior sequencing (Social-seq)-an occlusion-robust 3D pose estimation and unsupervised clustering pipeline-we found that nucleus accumbens (NAc) do...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-11-13 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Motor imagery (MI) can support self-initiated control, whereas brain-computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspend...

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Optogenetics in neurobiology and neurodegenerative disorders: from neural circuit mechanisms to precision neuromodulation

V Singaravel 2026-11-13 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: Neurodegenerative disorders, including Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis, multiple sclerosis and stroke, are characterized by progressive neuronal dysfunction, synaptic impairment and neural circuit degeneration, leading to severe cognitive...

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Brain-computer interfaces for neurodegenerative diseases: A comprehensive review of closed-loop systems from diagnosis to neural rehabilitation

Yadong Guo 2026-11-13 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) technology serves as an important bridge between the nervous system and external devices and has shown considerable promise for the diagnosis and treatment of neurodegenerative diseases, whose global burden continues to increase: in 2021, more than 57 million people we...

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Psychological Predictors of Pilot Performance: An Integrative Review of Proximal and Distal Performance Criteria

Xiang Yao 2026-11-13 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Research on pilots' psychological attributes and performance spans diverse predictors and performance criteria, but the findings remain fragmented. This integrative review synthesizes 60 empirical studies using a predictor-criterion framework that distinguishes cognitive abilities and behavioral dis...

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Cortical Network Alterations and Treatment-Induced Reorganization in Panic Disorder: A Longitudinal High-Density EEG Study

Yuan Yang 2026-11-13 1 min 33 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: PD is characterized by frequency-specific cortical network disruptions that show dynamic reorganization following treatment. These findings highlight the potential of EEG-derived network features as candidate markers for PD identification and treatment monitoring.

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A User-Function Taxonomy for Brain-Computer Interfaces: From Task Definition to Evidence

Xiaorong Gao 2026-11-13 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) studies are commonly compared by electrode, signal, paradigm, device, or clinical purpose. These descriptors do not identify the user-level function implemented in a task, although that function determines the appropriate endpoint. This article proposes the Type-Wield-...

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GB-algebraic class

Jiangang Tang 2026-11-13 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: The purpose of this article is to study the relationships among the BCI-algebraic class, the GB-algebraic class, and the class of algebras of type (2,0). The results indicate that: (1) The BCH-algebraic class and the GB-algebraic class are two distinct algebraic classes, both of which contain the BC...

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Saturday Daily Thread: Resource Request and Sharing! Daily Thread

/u/ AutoModerator 2026-11-14 1 min 180 words REDDIT PYTHON

Summary: <!-- SC_OFF --> <h1>Weekly Thread: Resource Request and Sharing 📚</h1> Stumbled upon a useful Python resource? Or are you looking for a guide on a specific topic? Welcome to the Resource Request and Sharing thread! <h2>How it Works:</h2> Request: Can't...

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https://www.reddit.com/r/Python/comments/1xvnujf/saturday_daily_thread_resource_request_and/

All US airports are currently ground stopped

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FAA issues nationwide ground stop due to "extraordinary circumstances"

👤 owenshen24 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

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Live airband on all frequencies from KORS. Victoria, CA also ground stopped

👤 mmaunder 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

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Lying to Zed to get its remote server running on FreeBSD

👤 arch1e 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

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How Not to Build a Music Player

👤 vinhnx 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

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Terminal borders that don't cost a cell

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What Happened to 3Com?

👤 gvieri 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

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Anthropic Upgrades Claude Code Projects to Run Parallel AI Workers Hands-Free

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Amtrak booking system experiencing major issues amid nationwide FAA ground stop

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> **Summary:** Article URL: https://www.fox5dc.com/news/amtrak-outage-ticket-booking-faa-ground-stop-november-13-2026 Comments URL: <a href="https://news.ycombinator.com/item?id=50492315">https://news....

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Hawaii DOT: All West Coast flights halted by SpaceX debris after FAA ground stop

👤 taylorfinley 📅 2026-11-14 📖 1 min 📝 13 words HACKER NEWS

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There are 13 ways to analyse a replication study, but only one of them is coherent.

noreply@blogger.com (Daniel Lakens)
2026-05-03 6 min 1361 words

TWENTY PERCENT STATISTICIAN

Summary: Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

Adriel Carridice
2026-06-22 1 min 0 words BRAIN

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https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

👤 Adriel Carridice 📅 2026-09-23 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

Naofucong mitigates amyloid-β deposition and cognitive deficits by enhancing insulin-degrading enzyme expression and inhibiting BACE through MAPK pathway inhibition

1 min 19 words BRAIN RESEARCH

Summary: Publication date: 1 February 2027 Source: Brain Research, Volume 1896 Author(s): Yue Tian, Mei Ma, Ruiying Yin, Guangchan Jing, Mengren Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004956?dgcid=rss_sd_all

Atlas-assisted segment anything model for fetal brain MRI (FeTal-SAM)

📖 1 min 📝 21 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Qi Zeng, Weide Liu, Bo Li, Ryne Didier, P. Ellen Grant, Davood Karimi

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006105?dgcid=rss_sd_all

Robust joint estimation of regional brain activity and inter-regional source covariance via full-structure covariance learning

📖 1 min 📝 21 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Chang Cai, Wenjun Deng, Kunpeng Zhao, Hepeng Ke, Wei Wu, Srikantan S. Nagarajan

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https://www.sciencedirect.com/science/article/pii/S1053811926006312?dgcid=rss_sd_all

Default mode network mechanisms of trait rumination in treatment-resistant depression and rTMS: evidence from source-localized EEG

📖 1 min 📝 26 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Stephanie A. Chu, Thomas E. Valles, Cole Citrenbaum, Michael K. Leuchter, Mohamad Shamas, Andrew F. Leuchter, Juliana Corlier

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926005999?dgcid=rss_sd_all

Audio large language models align with human neural responses during spoken-language comprehension

📖 1 min 📝 14 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Shiyao Zhu, Ao Chen, Yanlu Xie

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006580?dgcid=rss_sd_all

Neural dynamics track the approach to physical exhaustion

Avancini, C., Arco, J. E., Alameda, C., Bekinschtein, T. A., Sanabria, D., Ciria, L. F.

2026-11-13

1 min | 123 words

BIORXIV NEUROSCIENCE

Summary: Physical exhaustion has long been described through its cardiorespiratory and metabolic cost, while the neural dynamics of its approach remain largely unexplored. Here we recorded high-density electroencephalography in 28 trained cyclists during a maximal incremental test to characterise how neural ...

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https://www.biorxiv.org/content/10.64898/2026.11.12.766118v1?rss=1

Zdhhc17/Hip14 functions in neurons to control behavior via DHHC-dependent and independent mechanisms

Cheng, Y., Santistevan, N. J., Fink, K., Jacobson, S., Mosimann, C., Burger, A., Nelson, J. C.

2026-11-13 | 1 min | 218 words

BIORXIV NEUROSCIENCE

Summary: To navigate a complex sensory environment, animals filter out inconsequential sensory inputs and habituate to familiar stimuli. By establishing innate response thresholds and modifying them through habituation, animals can focus on key sensory stimuli while ignoring irrelevant ones. Huntingtin-inter...

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https://www.biorxiv.org/content/10.64898/2026.11.12.766291v1?rss=1

A Variational Principle for the Emergence of Neural Geometry Underlying Bayesian-Like Behavior

Serrano-Fernandez, L., Parga, N.
2026-11-13 1 min 173 words BIORXIV NEUROSCIENCE

Summary: Neural population geometry constrains how task variables are represented and read out into behavior. We propose a variational principle in which learning balances task accuracy and representational smoothness under stimulus-dependent weighting. This balance links stimulus statistics to the shape of ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765947v1?rss=1

Satellite glial translatome reprograms to favor mitochondrial and vesicular signaling under pelvic pain conditions

Yesupatham, S. K., Granley, A., Xie, A. X.
2026-11-13 1 min 150 words BIORXIV NEUROSCIENCE

Summary: Sensory satellite glial cells actively contribute to inflammatory and neuropathic pain, but their roles in pelvic pain remain unclear. This study examined how the active translating mRNAs in sensory SGCs change during pelvic pain symptoms using translating ribosome affinity purification followed by ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.766000v1?rss=1

Amyloidosis and tauopathy progressively disrupt near-critical dynamics in selectively vulnerable circuits

McGregor, J. N., Fosque, L. J., Brunwasser, S. J., Farris, C., Higashikubo, B. T., Berriman-Rozen, Z. D., Elmore, H., Kest, K., Whitesell, J. D., Harris, J. A., Shew, W. L., Holtzman, D. M., Hengen, K. B.

2026-11-13 1 min 282 words BIORXIV NEUROSCIENCE

Summary: Mechanistically distinct neurodegenerative diseases result in similar cognitive deficits, and pathology is often concentrated in a set of circuits known to be selectively vulnerable. Selective vulnerability is an intrinsic property of a circuit: in APP/PS1 mice, where plaques are global, functional ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.11.766020v1?rss=1

Electrophysiology of multisite direct electrical stimulation in the human brain

Sehgal, D., Paulk, A. C., Hadar, P. N., Peralta, J. D., Salami, P., Rolston, J. D., Richardson, R. M., Cash, S. S., Zelmann, R.

2026-11-13 1 min 150 words BIORXIV NEUROSCIENCE

Summary: Human cognition and neurological disorders reflect properties of distributed networks, yet how best to engage these networks therapeutically remains unknown. Multisite stimulation is a principled strategy for engaging brain networks, but a key translational barrier is understanding how a network com...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.11.765993v1?rss=1

Repeated self-administered EEG recordings enable longitudinal measurement of brain activity at home

van Moorselaar, D., Kock, R., Theeuwes, J.

2026-11-13 1 min 193 words

BIORXIV NEUROSCIENCE

Summary: Mobile electroencephalography (EEG) offers new opportunities to investigate brain activity outside the laboratory, but the quality and reliability of self-administered recordings remain important concerns. Here, we evaluated whether repeated at-home recordings using a four-channel consumer EEG devic...

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https://www.biorxiv.org/content/10.64898/2026.11.12.766143v1?rss=1

Nucleus reuniens gates selective hippocampal-prefrontal interactions during NREM sleep

Shao, S., Ferrero Lopez, J., Kawano, T., Bhagvat, C., Khodagholy, D., Gelinas, J.

2026-11-13

1 min 141 words

BIORXIV NEUROSCIENCE

Summary: Memory consolidation requires functional synchrony between the hippocampus (HC) and the medial prefrontal cortex (mPFC). The nucleus reuniens (RE) is a structural and functional hub between HC and mPFC. Here, we used in vivo electrophysiology to monitor the hippocampal-thalamic-prefrontal network in...

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https://www.biorxiv.org/content/10.64898/2026.11.11.765971v1?rss=1

Inactivation of tissue-nonspecific alkaline phosphatase reduces phosphatidylcholine amounts in the brain of infant mice

👤 Nowak, L. G., Marie, G., Balayssac, S., Delos, O., Mornet, E., Gilard, V., Bertrand-Michel, J., Fonta, C.

📅 2026-11-13 📖 1 min 📝 206 words BIORXIV NEUROSCIENCE

Summary: Hypophosphatasia is a rare disease caused by dysfunctions of tissue-nonspecific alkaline phosphatase (TNAP). TNAP is expressed in numerous tissues, notably the brain, and patients with hypophosphatasia exhibit neurological symptoms, including epilepsy in the most severe forms of the disease. TNAP kn...

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https://www.biorxiv.org/content/10.64898/2026.11.12.766160v1?rss=1

Microglial Stim1/Stim2 deletion reduces calcium signaling but promotes infarct expansion after ischemic stroke

👤 Sharifi, K. A., Norat, P., Jannatifar, N., Wang, Z., Vergeres, C., Padigala, M., Sluzewski, M. F., Liu, L., Zhao, A., Hoch, M. R., Le, H., Jansen, M. A., Kellogg, R. T., Kuan, C.-Y., Miller, G. W., Tvrdik, P.

📅 2026-11-13 📖 1 min 📝 225 words BIORXIV NEUROSCIENCE

Summary: Microglia respond to ischemic stroke with calcium transients carried by calcium release-activated calcium (CRAC) channels, gated by the endoplasmic reticulum sensors Stim1 and Stim2. Since CRAC signaling drives pro-inflammatory responses in immune cells, blocking it in microglia should protect the i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.12.766275v1?rss=1

Intrinsic neural dynamics reveal a body-part-specific topography in mouse secondary motor cortex

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78900-3

Failure of TSPO-dependent metabolic compensation marks the loss of mitochondrial bioenergetic adaptation in parkin-deficient neurons

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41467-026-79500-x

The neuroscience of delirium: a perfect storm for acute brain dysfunction

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Mobile Health Apps for Individuals Who Are Blind or Have Low Vision: Systematic Review of Evidence, Accessibility, and App Quality

👤 Jordi Miró 📅 2026-11-12 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: The current landscape of mobile health apps for individuals who are blind or have low vision remains limited, uneven in accessibility and quality, and supported by a limited and heterogeneous evidence base. The absence of identified apps targeting psychological functioning represents a ...

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DPCoOp: Dual positive context optimization for multi-label recognition with limited annotations

👤 Yingchun Kuang 📅 2026-11-12 📖 1 min 📝 67 words LOW VISION

Summary: Multi-label recognition with limited annotations has attracted significant attention due to its close alignment with real-world needs. Vision-language pretraining models like CLIP have effectively mitigated inadequate feature learning from scarce annotations, making them a central research focus in ...

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Automated detection of blink reflexes evoked by optogenetic stimulation of TRPV1-lineage corneal afferents in transgenic mice

👤 Carl Y Saab 📅 2026-11-12 📖 1 min 📝 28 words LOW VISION

Summary: CONCLUSIONS: This video-based method enables accurate, high-throughput, and observer-independent empirical assessment of stimulus-evoked blink reflexes, providing a quantitative platform for investigating corneal sensory responses relevant to ocular nociception.

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Step change in glaucoma polygenic risk score performance enables clinical utility

👤 Jamie E Craig 📅 2026-11-12 📖 1 min 📝 58 words LOW VISION

Summary: CONCLUSIONS: This enhanced PRS, available for clinical use through an accredited genotyping workflow, enables effective risk stratification for OAG across diverse ancestries, facilitating earlier identification and treatment of high-risk individuals while supporting resource rationalisation for thos...

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A robust framework for cross-level spatial-frequency collaborative recognition in plant disease images

Haiming Lan 2026-11-12 1 min 61 words LOW VISION

Summary: Plant disease images commonly suffer from pixel confusion, texture blurring and semantic ambiguity caused by cluttered backgrounds, noise interference and diverse lesion morphologies. These intra- and inter-class disturbances break the representation balance between global context and local details,...

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Creating perceptually uniform circular shape spaces: An automated method for asymmetric naturalistic shapes

Yaniv Morgenstern 2026-11-12 1 min 63 words LOW VISION

Summary: Shape processing is crucial for higher-level visual functions such as object recognition. Yet, isolating high-level vision from low-level shape influences remains challenging. Typically, studies use simple, symmetric stimuli and labor-intensive psychophysical measurements to control for perceptual s...

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Genetic parameters and genomic prediction accuracy for piling behaviour using individual-level computer-vision data from ArUco-marked laying hens

👤 P Bijma 📅 2026-11-12 📖 1 min 📝 69 words LOW VISION

Summary: CONCLUSIONS: This study provides the first evidence that piling behaviour in laying hens is a heritable trait. The significant genetic variance and predicted response to selection across traits confirm the potential for piling mitigation through genetic selection. A practical strategy for breeding p...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

Joseph Paul Nemargut

2026-11-12 1 min 69 words LOW VISION

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Computer vision in precision poultry farming: evaluation of YOLO and transformer-based object detectors for dead broiler detection in farm inspection images

👤 Petr Bartoš 📅 2026-11-13 📖 1 min 📝 71 words LOW VISION

Summary: Broiler mortality inspection is an important component of poultry farm management. This study presents a pilot technical feasibility evaluation of recent YOLO and transformer-based object detection models for detecting dead broilers in event-enriched farm inspection images. The dataset was collected...

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Uptake of and adherence to diabetic retinopathy screening in an opportunistic setting: a quasi-experimental cohort study from Upper Austria

👤 Josef Huemer 📅 2026-11-13 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSION: Although DMPs based on an opportunistic model improve overall adherence to screening guidelines, they do not achieve the same level of participation in retinopathy screening as systematic programmes in other countries.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

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Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

> **Summary:** This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

👤 Joseph Paul Nemargut 📅 2026-11-12 📖 1 min 📝 69 words BRAILLE

> **Summary:** CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

👤 Sevda Uzun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Prefrontal transcranial direct current stimulation combined with cognitive training reduces anxiety, daytime sleepiness, and cognitive complaints in women after COVID-19: A randomized, double-blind, sham-controlled trial

👤 Guilherme Augusto Santos Bueno
📅 2026-11-12 📖 1 min 📝 37 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Adding left-prefrontal tDCS to cognitive training reduced anxiety and daytime sleepiness and improved cognition and psychomotor speed in women with post-COVID cognitive complaints, with parallel changes in prefrontal EEG markers. Confirmation in larger samples is needed.

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Home-based transcranial direct current stimulation and exercise for pain in people with knee osteoarthritis: A mixed-methods pilot randomised controlled trial

👤 Rocco Cavaleri 📅 2026-11-12 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: This mixed-methods pilot trial evaluated the feasibility and patient perceptions of home-based transcranial direct current stimulation (tDCS) combined with exercise for pain in people with knee osteoarthritis. Twenty-four participants were allocated to active or sham home-based tDCS plus exercise. P...

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Effects of high-definition transcranial direct current stimulation on muscle strength, proprioception, cutaneous sensitivity and their relationship with balance in older adults with a history of falls

👤 Wei Sun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: A 4-week HD-tDCS intervention significantly improves static and dynamic balance, hip strength, and ankle inversion proprioceptive in older adults with a history of falls. Improved static balance is positively correlated with increased hip abduction strength.

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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Acute aerobic exercise and orbitofrontal cortex reactivity during positive emotion processing in obesity: An exploratory fNIRS study

👤 Jun Yan 📅 2026-11-12 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSION: Obese students showed lower baseline PA and right OFC activation during active positive elaboration; post-exercise PA and right OFC activation were higher in the same context. These single-arm associations cannot be attributed to exercise; right OFC is a preliminary trial signal, dlPFC i...

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Motor imagery enhances swallowing motor cortex excitability and activates sensorimotor regions: a TMS and fNIRS study

👤 Yue Lan 📅 2026-11-12 📖 1 min 📝 51 words FNIRS

Summary: CONCLUSION: MI enhances the excitability of the swallowing motor cortex and activates the key sensorimotor cortical areas governed ME of swallowing. MI shows greater superiority over AO and may become a promising effective rehabilitation strategy for neurogenic dysphagia. Trial registration Chinese ...

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A translational platform for brain-computer interfaces and adaptive neuromodulation: Technical characterization, long-term validation, and implementation of the CorTec brain interchange-BCI2000 ecosystem

👤 Kai J Miller
📅 2026-11-12 📖 1 min 📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Adaptive neuromodulation systems and implantable brain-computer interfaces (BCIs) are promising therapies for neurological and psychiatric disorders. However, their broader translation into research and clinical practice remains limited by technological complexity, restricted access to implantable r...

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Low-amplitude intracortical microstimulation produces short-latency modulation throughout the cortical reach-to-grasp network

Marc H Schieber 2026-11-12 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Intracortical microstimulation (ICMS) can be used to deliver information to the cortex from a bidirectional brain-machine interface (BMI), thereby improving control of neuroprosthetic devices. Primary somatosensory cortex (S1) is a common choice for delivering information, but S1-ICMS can...

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Hierarchical causal construction for semantic-granularity-aligned counterfactual medical image generation

Yifan Chen 2026-11-12 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Counterfactual image generation seeks to answer how a medical image should change when a clinical semantic factor is altered, and is important for model interpretation, lesion localization, and controllable image editing. This work addresses the semantic granularity mismatch in existing counterfactu...

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Sub-second dopamine reinforcement shapes juvenile social play

Ying Li 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Juvenile social play is critical for neurodevelopment, yet the neural mechanisms organizing moment-to-moment interactions remain elusive. Using social behavior sequencing (Social-seq)-an occlusion-robust 3D pose estimation and unsupervised clustering pipeline-we found that nucleus accumbens (NAc) do...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-11-13 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Motor imagery (MI) can support self-initiated control, whereas brain-computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspend...

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Optogenetics in neurobiology and neurodegenerative disorders: from neural circuit mechanisms to precision neuromodulation

👤 V Singaravel 📅 2026-11-13 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: Neurodegenerative disorders, including Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis, multiple sclerosis and stroke, are characterized by progressive neuronal dysfunction, synaptic impairment and neural circuit degeneration, leading to severe cognitive...

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Brain-computer interfaces for neurodegenerative diseases: A comprehensive review of closed-loop systems from diagnosis to neural rehabilitation

👤 Yadong Guo 📅 2026-11-13 📖 1 min 📝 70 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) technology serves as an important bridge between the nervous system and external devices and has shown considerable promise for the diagnosis and treatment of neurodegenerative diseases, whose global burden continues to increase: in 2021, more than 57 million people we...

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Psychological Predictors of Pilot Performance: An Integrative Review of Proximal and Distal Performance Criteria

👤 Xiang Yao 📅 2026-11-13 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Research on pilots' psychological attributes and performance spans diverse predictors and performance criteria, but the findings remain fragmented. This integrative review synthesizes 60 empirical studies using a predictor-criterion framework that distinguishes cognitive abilities and behavioral dis...

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Cortical Network Alterations and Treatment-Induced Reorganization in Panic Disorder: A Longitudinal High-Density EEG Study

👤 Yuan Yang 📅 2026-11-13 📖 1 min 📝 33 words BRAIN COMPUTER INTERFACE

> **Summary:** CONCLUSIONS: PD is characterized by frequency-specific cortical network disruptions that show dynamic reorganization following treatment. These findings highlight the potential of EEG-derived network features as candidate markers for PD identification and treatment monitoring.

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Eventually LLMs will write what will become industry standard python.

👤 /u/DistributionBrief644 📅 2026-11-13 📖 1 min 📝 71 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <p>LLMs are still outputting code that is too sloppy for some engineers. However, there will come a point in the near future where a model can write a "standard" style of code. Style of code is something most python developers recognise, and i wonder if that ...

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Microsoft outage gives customers a crash course in dependence on Redmond

👤 nephihaha 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theregister.com/saas/2026/11/13/microsoft-outage-gives-customers-a-crash-course-in-dependence-on-redmond/5309805 Comments URL:...

🔗 **Read full article:**
https://www.theregister.com/saas/2026/11/13/microsoft-outage-gives-customers-a-crash-course-in-dependence-on-redmond/5309805

Models can now transmit viruses in weight-space

👤 mkglees 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.alphaxiv.org/abs/2611.weight-space-misfolding Comments URL: https://news.ycombinator.com/item?id=50487762 Points: 99 # ...

🔗 **Read full article:**
https://www.alphaxiv.org/abs/2611.weight-space-misfolding

$11.4M in Nvidia hardware vanishes during shipment from CA to TX

👤 crescit_eundo 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.freightwaves.com/news/11-4-million-in-nvidia-hardware-vanishes-during-shipment-from-california-to-texas Comments URL: <a href="https://...

🔗 **Read full article:**
https://www.freightwaves.com/news/11-4-million-in-nvidia-hardware-vanishes-during-shipment-from-california-to-texas

FM-1 Workbench

👤 simonebrunozzi 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://open.m-vave.com/fm1-workbench Comments URL: https://news.ycombinator.com/item?id=50488325 Points: 86 # Comments: 48

🔗 **Read full article:**
https://open.m-vave.com/fm1-workbench

Perplexity Decider v1.3: Training, eval, and use cases

👤 gmays 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.perplexity.ai/hub/blog/perplexity-decider Comments URL: https://news.ycombinator.com/item?id=50488464 Points: 35 # Comments...

🔗 **Read full article:**
https://www.perplexity.ai/hub/blog/perplexity-decider

Found: A protein that drives ageing without unleashing inflammation

👤 gumby 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.nature.com/articles/d41586-026-03638-3 Comments URL: https://news.ycombinator.com/item?id=50488470 Points: 27 # Comments: 3</p...

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03638-3

Technology as a Search Problem

👤 dverlaeckt80 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.construction-physics.com/p/technology-as-a-search-problem Comments URL: https://news.ycombinator.com/item?id=50488490 ...

🔗 **Read full article:**
https://www.construction-physics.com/p/technology-as-a-search-problem

Someone's Impersonating This Blog

👤 speckx 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://mattbee.mataroa.blog/p/someones-impersonating-this-blog/ Comments URL: https://news.ycombinator.com/item?id=50488664 Points...

🔗 **Read full article:**
https://mattbee.mataroa.blog/p/someones-impersonating-this-blog/

No AI Slop: A Manifesto

👤 bankombinator 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.slopmanifesto.com/ Comments URL: https://news.ycombinator.com/item?id=50488769 Points: 54 # Comments: 26

🔗 **Read full article:**
https://www.slopmanifesto.com/

Big tech is here to let us down

👤 speckx 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.jsweeting.me/big-tech-is-here-to-let-us-down Comments URL: https://news.ycombinator.com/item?id=50489083 Points: 26 # ...

🔗 **Read full article:**
https://blog.jsweeting.me/big-tech-is-here-to-let-us-down

Type S and M errors as a “rhetorical tool”

noreply@blogger.com (Daniel Lakens)
2025-09-28 17 min 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

noreply@blogger.com (Daniel Lakens)
2025-10-29 27 min 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

2026 Adcom Petition is Open

👤 Deidre Artis 📅 2026-09-11 📖 1 min 📝 14 words EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Subacute administration of human umbilical cord-derived mesenchymal stem cells is associated with neurobehavioral improvement, myeloid immune modulation, and altered HMGB1 dynamics after experimental ischemic stroke

📖 1 min 📝 21 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Yuki Takeda, Yoji Kuramoto, Shoichiro Tsuji, Nobutaka Doe, Kenichi Yamahara, Shinichi Yoshimura

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004798?dgcid=rss_sd_all

Prenatal cocaine exposure alters synaptic plasticity of hippocampal CA1 pyramidal neurons in postnatal rats

📖 1 min 📝 29 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jian Fu, Cheng-Yan Sun, Qing-Yun Yang, Long Jiang, Yong Tang, Jie Shen, Shun-Nan Ge, Xin Wang, Xue-Lian Wang, Liang Qu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004907?dgcid=rss_sd_all

Enriched environment protects from high-fat diet-induced obesity but not from insulin resistance, glucose intolerance, and metabolic deterioration in the brain

📖 1 min 📝 41 words BRAIN RESEARCH

Summary: Publication date: 1 February 2027 Source: Brain Research, Volume 1896 Author(s): Anne Ladwig, Anna Lena Cremer, Johannes Romer, Valentina Zimmermann, Claus Brandt, Stefan Blaschke, Claudia Orschel, Charlotte Orschel, Marvin Schäfer, Maria A. Rueger, Klarissa H. Stürner, Gereon ...

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004853?dgcid=rss_sd_all

Mechanism by which metformin alleviates neuralgia in the spinal cord of diabetic rats

📖 1 min 📝 24 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Xiang Chen, Meng-Jie Tang, Ge Song, Ya Shen, Li-Min Shen, Yi-Feng Sha, Ye-Zi Sun, Xiao-Jun Cao

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S030645222600730X?dgcid=rss_sd_all

Studying the pharmacological interaction of temozolomide chemotherapy with different antidepressant treatments in a rat glioblastoma model

📖 1 min 📝 17 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Laura Gálvez-Melero, Julia Bihorac, Martin Hadamitzky, M. Julia García-Fuster

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007323?dgcid=rss_sd_all

A spatially continuous cell-type atlas reveals multiscale organization of the human cortex

👤 Lei, W., Tian, X., Dong, C., Lou, J., Zhao, X., Dong, X., Du, L., Yang, R., Hong, Y., Yu, Y., Yang, C., Qian, L., Liu, S., Li, A., Wang, M., Liu, B.

📅 2026-11-12 📖 1 min 📝 190 words BIORXIV NEUROSCIENCE

Summary: Neuroimaging has increasingly revealed fine-grained spatial heterogeneity within the human cortex, uncovering regional differentiation beyond conventional cortical divisions. However, understanding the cellular architecture underlying these imaging features requires spatially continuous cell-type re...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765231v1?rss=1

A cortically gated sensory reference to the basal ganglia guides vocal reinforcement learning in songbirds

Agarwal, A., Scharff, C., Vallentin, D.

2026-11-13 1 min 192 words

BIORXIV NEUROSCIENCE

Summary: Songbirds learn to sing by copying a memorized tutor song. Reinforcement learning models of vocal imitation propose that a dopaminergic signal strengthens the cortico basal ganglia synapses whose activity coincides with similarity to the tutor like song. To do so, the circuit must access a represent...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765895v1?rss=1

Local consequences of FMRP loss: cell type-specific mRNA and protein profiles of Fmr1-KO synapses

Desch, K., Aghabashlou Saisan, S., tom Dieck, S., Fuerst, N., Nassim-Assir, B., Langer, J. D., Juengling, M., Kaulich, E., Schuman, E. M.

2026-11-13 1 min 248 words BIORXIV NEUROSCIENCE

Summary: Brain function relies on billions of neurons interconnected by synapses with unique molecular composition specialized for cell type, brain region, and their role in circuit dynamics. Pathological synaptic dysfunctions - so-called synaptopathies - underlie many neurodevelopmental disorders (NDDs), in...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765892v1?rss=1

Oscillatory Entrainment Guides Temporal Prediction in Non-Deterministic Continuous Environments, Independent of Bayesian Sequential Learning

👤 Hosseini, E., Breska, A. 📅 2026-11-13 📖 1 min 📝 148 words BIORXIV NEUROSCIENCE

Summary: Predicting future event timing is essential for human interaction with our environment, and has been established in stimulus streams containing deterministic regularities. However, real-world streams often lack such regularities, leaving it unclear whether and how temporal predictions operate. Here ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765929v1?rss=1

Channel expression aligns with dendritic conductance load in the reinforcing, not compensatory, direction in cortical GABAergic neurons

👤 Cirrincione, G., Grassia, F. 📅 2026-11-13 📖 1 min 📝 181 words BIORXIV NEUROSCIENCE

Summary: Neurons could offset the conductance load imposed by their dendrites through channel expression (compensation), or channel expression could move with that load (reinforcement); which arrangement holds in adult cortex is unknown. Writing rheobase as the product of input conductance and an effective-v...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765852v1?rss=1

New disease models for investigating SynGAP1-related disorder (SRD) caused by a common heterozygous nonsense SYNGAP1 mutation

👤 von Sivers, J., Unger, C., Bauersachs, D., Kupryianchyk-Schultz, I., Gupta, T., Proce, R. O., Rybak-Wolf, A., Rademacher, N., Schmitz, D., Schuelke, M., Hoernberg, H., Kuehn, R., Shoichet, S. A.

📅 2026-11-13 📖 1 min 📝 238 words BIORXIV NEUROSCIENCE

Summary: Mutations in the SYNGAP1 gene cause SynGAP1-related disorder (SRD), a severe neurodevelopmental disorder marked by intellectual disability, epilepsy, ataxia, autism spectrum disorder, and global developmental delay. Studies in mouse models of SRD show that restoration of SynGAP1 expression in adult ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765833v1?rss=1

A neural circuit for stereopsis in the fly

👤 Mearns, D. S., Park, Y., Shin, I., Murthy, M.

📅 2026-11-13 📖 1 min 📝 286 words

BIORXIV NEUROSCIENCE

Summary: Stereopsis involves combining different views from the two eyes to infer depth, an ability that has intrigued neuroscientists for almost two centuries (Wheatstone, 1838). It enables insects such as the praying mantis to estimate the distance of prey (Nityananda et al., 2018), or primates to fixate o...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.11.765991v1?rss=1

The aging human brain exhibits reduced hemodynamic and cerebrospinal fluid flow waves during sleep

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s43587-026-01270-4

Enhancer-targeted CRISPR-A rescues haploinsufficiency and mutant phenotypes in organoid models of autism

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02504-9

High resolution multi-depth quantification of the retinal nerve fiber layer

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78810-w

Glutamate-binding protein from Corynebacterium glutamicum as fluorescence probe for glutamate detection in artificial cerebrospinal fluid

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-80347-x

Transform-based audio processing for neuromorphic cochlear implants

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76746-9

Spectral moment embedding: a matrix-moment generalisation of characteristic response analysis for covariance-based signal classification

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-67565-z

Aging disrupts sleep-driven brain fluid flow

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s43587-026-01296-8

Vagal afferent activation induces a dissociated NREM sleep-like state with wake-related locus coeruleus activity

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79135-y

Enhancer-targeted CRISPR-A rescues haploinsufficiency and mutant phenotypes in organoid models of autism

Daniel H. Geschwind

2026-11-13 | 1 min | 39 words | NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 13 November 2026; doi:10.1038/s41593-026-02504-9 CRISPR activation targeted to enhancers of autism risk genes CHD8 and SCN2A selectively increased expression and reversed mutant phenotypes i...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02504-9

The decision to stop “counting” explains numerical underproduction in primates

Laura E. SeidlerLena L. JannaschStephanie WestendorffAndreas Niederahttps://ror.org/03a1kwz48Animal Physiology Unit, Institute of Neurobiology, University of Tübingen, Tübingen 72076, Germany

2026-11-09 | 1 min | 51 words | PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 46, November 2026.
SignificanceProducing a specific number of actions is a fundamental component of counting, yet little is known about how the brain determines when a counting sequence should end. We show that two rhesus macaq...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2620933123?af=R

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

From biomechanical data to robotic execution: design and control of an intelligent Tuina system

Wei Yao 2026-11-02 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: IntroductionTraditional Chinese medicine (TCM) Tuina (massage) relies heavily on practitioners' physical stamina and subjective experience, leading to operational variability that hinders therapeutic standardization and scalable rehabilitation service delivery. To address this limitation, this study...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1937295

Path planning algorithm based on pulse-coupled neural network with dual constraints of heading-deviation and node penalty

Xiaoyan Wang 2026-11-05 1 min 321 words FRONTIERS NEUROROBOTICS

Summary: Path planning for mobile robots is a crucial aspect of robot navigation. Traditional search algorithms are limited by heuristic design and search strategies, often struggling to balance search efficiency and path quality in intricate obstacle environments. The Pulse-Coupled Neural Network (PCNN) off...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1907089

Multimodal fMRI and DTI correlates of affective symptoms in diarrhea-predominant irritable bowel syndrome: a subtype-focused neuroimaging study

Lihao Xiao 2026-11-13 1 min 288 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo investigate brain structural and functional alterations in diarrhea-predominant irritable bowel syndrome (IBS-D) and their associations with affective symptoms using multimodal magnetic resonance imaging (MRI).Methods53 IBS-D patients and 55 healthy controls (HCs) underwent resting-state...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2000144

Co-adaptive learning and domain adaptation bridge the translational gap in neuroprosthetic control: a review

Ishika K. Kanakath 2026-11-13 1 min 185 words FRONTIERS NEUROSCIENCE

Summary: Myoelectric and neural prosthetic devices frequently fail in daily clinical use despite laboratory success, a persistent translational gap driven by domain shift, the non-stationary nature of biological signals arising from neural plasticity, motor learning, muscle fatigue, and electrode displacemen...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1974861

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113095004&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113095004&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113095004&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113095004&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113095004&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Seeing Text Clearly in the Dark: Unifying Localization and Enhancement via Deep Morphological Synergy

Rushi Lan 2026-11-12 1 min 67 words LOW VISION

Summary: Low-light text localization, an interdisciplinary field, has gained significant attention recently. The goal is to provide enhanced text locational information beneficial for both machine and human vision. Existing one-stage and two-stage methods either rely on single detectors specially designed fo...

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RPDL-YOLO: An efficient YOLOv8 framework for small bird detection in airport airside environments

👤 Heng Yang 📅 2026-11-12 📖 1 min 📝 63 words LOW VISION

Summary: Bird intrusions into airport security zones pose significant threats to aviation safety. However, the small size, rapid movement, and irregular flight patterns of birds often result in missed detections and false alarms in vision-based surveillance systems, especially in complex open-sky environment...

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Mobile Health Apps for Individuals Who Are Blind or Have Low Vision: Systematic Review of Evidence, Accessibility, and App Quality

👤 Jordi Miró 📅 2026-11-12 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: The current landscape of mobile health apps for individuals who are blind or have low vision remains limited, uneven in accessibility and quality, and supported by a limited and heterogeneous evidence base. The absence of identified apps targeting psychological functioning represents a ...

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DPCoOp: Dual positive context optimization for multi-label recognition with limited annotations

👤 Yingchun Kuang 📅 2026-11-12 📖 1 min 📝 67 words LOW VISION

Summary: Multi-label recognition with limited annotations has attracted significant attention due to its close alignment with real-world needs. Vision-language pretraining models like CLIP have effectively mitigated inadequate feature learning from scarce annotations, making them a central research focus in ...

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Automated detection of blink reflexes evoked by optogenetic stimulation of TRPV1-lineage corneal afferents in transgenic mice

👤 Carl Y Saab 📅 2026-11-12 📖 1 min 📝 28 words LOW VISION

Summary: CONCLUSIONS: This video-based method enables accurate, high-throughput, and observer-independent empirical assessment of stimulus-evoked blink reflexes, providing a quantitative platform for investigating corneal sensory responses relevant to ocular nociception.

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Step change in glaucoma polygenic risk score performance enables clinical utility

Jamie E Craig 2026-11-12 1 min 58 words LOW VISION

Summary: CONCLUSIONS: This enhanced PRS, available for clinical use through an accredited genotyping workflow, enables effective risk stratification for OAG across diverse ancestries, facilitating earlier identification and treatment of high-risk individuals while supporting resource rationalisation for thos...

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A robust framework for cross-level spatial-frequency collaborative recognition in plant disease images

Haiming Lan 2026-11-12 1 min 61 words LOW VISION

Summary: Plant disease images commonly suffer from pixel confusion, texture blurring and semantic ambiguity caused by cluttered backgrounds, noise interference and diverse lesion morphologies. These intra- and inter-class disturbances break the representation balance between global context and local details,...

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Creating perceptually uniform circular shape spaces: An automated method for asymmetric naturalistic shapes

👤 Yaniv Morgenstern 📅 2026-11-12 📖 1 min 📝 63 words LOW VISION

Summary: Shape processing is crucial for higher-level visual functions such as object recognition. Yet, isolating high-level vision from low-level shape influences remains challenging. Typically, studies use simple, symmetric stimuli and labor-intensive psychophysical measurements to control for perceptual s...

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Genetic parameters and genomic prediction accuracy for piling behaviour using individual-level computer-vision data from ArUco-marked laying hens

👤 P Bijma 📅 2026-11-12 📖 1 min 📝 69 words LOW VISION

Summary: CONCLUSIONS: This study provides the first evidence that piling behaviour in laying hens is a heritable trait. The significant genetic variance and predicted response to selection across traits confirm the potential for piling mitigation through genetic selection. A practical strategy for breeding p...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

👤 Joseph Paul Nemargut
📅 2026-11-12 📖 1 min 📝 69 words

LOW VISION

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Impact of specialized mobility services on independent travel, economic outcomes, and well-being for people with low vision and blindness: an international survey

👤 Joseph Paul Nemargut 📅 2026-11-12 📖 1 min 📝 69 words BRAILLE

Summary: CONCLUSIONS: Specialized mobility instruction is associated with higher independence and well-being of PVI around the world. These services should be made more accessible to promote independent travel amongst PVI and potentially reduce the burden on caregivers and society. Attention should be given ...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

👤 Sevda Uzun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Prefrontal transcranial direct current stimulation combined with cognitive training reduces anxiety, daytime sleepiness, and cognitive complaints in women after COVID-19: A randomized, double-blind, sham-controlled trial

👤 Guilherme Augusto Santos Bueno
📅 2026-11-12 📖 1 min 📝 37 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Adding left-prefrontal tDCS to cognitive training reduced anxiety and daytime sleepiness and improved cognition and psychomotor speed in women with post-COVID cognitive complaints, with parallel changes in prefrontal EEG markers. Confirmation in larger samples is needed.

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Home-based transcranial direct current stimulation and exercise for pain in people with knee osteoarthritis: A mixed-methods pilot randomised controlled trial

👤 Rocco Cavaleri 📅 2026-11-12 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: This mixed-methods pilot trial evaluated the feasibility and patient perceptions of home-based transcranial direct current stimulation (tDCS) combined with exercise for pain in people with knee osteoarthritis. Twenty-four participants were allocated to active or sham home-based tDCS plus exercise. P...

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Effects of high-definition transcranial direct current stimulation on muscle strength, proprioception, cutaneous sensitivity and their relationship with balance in older adults with a history of falls

👤 Wei Sun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: A 4-week HD-tDCS intervention significantly improves static and dynamic balance, hip strength, and ankle inversion proprioceptive in older adults with a history of falls. Improved static balance is positively correlated with increased hip abduction strength.

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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Acute aerobic exercise and orbitofrontal cortex reactivity during positive emotion processing in obesity: An exploratory fNIRS study

👤 Jun Yan 📅 2026-11-12 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSION: Obese students showed lower baseline PA and right OFC activation during active positive elaboration; post-exercise PA and right OFC activation were higher in the same context. These single-arm associations cannot be attributed to exercise; right OFC is a preliminary trial signal, dlPFC i...

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Motor imagery enhances swallowing motor cortex excitability and activates sensorimotor regions: a TMS and fNIRS study

👤 Yue Lan 📅 2026-11-12 📖 1 min 📝 51 words FNIRS

Summary: CONCLUSION: MI enhances the excitability of the swallowing motor cortex and activates the key sensorimotor cortical areas governed ME of swallowing. MI shows greater superiority over AO and may become a promising effective rehabilitation strategy for neurogenic dysphagia. Trial registration Chinese ...

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Spinal cord-computer interfaces for neurorehabilitation: Technical challenges and clinical perspectives

Minfei Wu 2026-11-11 1 min 35 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: This Perspective aims to clarify the clinical and engineering scope of spinal cord-computer interfaces (SCCIs) in neurorehabilitation, distinguishing them from adjacent brain-computer interface, neuromodulation, and assistive-technology approaches while identifying the key barriers to tra...

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Clinical safety assessment of implanted brain-computer interfaces: current evidence and a pathway to standardization

Wen-Jun Tu 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Implanted brain-computer interfaces are beginning to move beyond short laboratory sessions into long-term and home use. The evidence used to judge clinical safety remains fragmented. This review examines penetrating intracortical, cortical-surface (including subpial) and cerebral-venous systems, wit...

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Characteristics of EEG Microstate Analysis of Power Grip and Precision Grip Hand Movements from Motor Planning to Execution

👤 Gang Wang
📅 2026-11-12 📖 1 min 📝 63 words

BRAIN COMPUTER INTERFACE

Summary: Hand movement is a fundamental daily motor behavior. Precision grip and power grip reflect distinct post-stroke recovery stages and neuroplasticity, making their neural mechanisms critical for motor control and neurorehabilitation. This study employed electroencephalographic (EEG) microstate analysi...

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Robotic Gait Multi-channel Mu Beta Neurofeedback Training in Paraplegics Enhanced Inter-regional Connectivity and Sensorimotor Functions

👤 Denis Delisle-Rodriguez
📅 2026-11-12 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Complete spinal cord injury (SCI) that severely affects somatosensory and motor functions, greatly challenges advanced neuro-technologies and therapies addressed to repair neural system pathways and recover lost functions. Non-invasive solutions have been explored, reaching encouraging results. This...

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A translational platform for brain-computer interfaces and adaptive neuromodulation: Technical characterization, long-term validation, and implementation of the CorTec brain interchange-BCI2000 ecosystem

👤 Kai J Miller
📅 2026-11-12 📖 1 min 📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Adaptive neuromodulation systems and implantable brain-computer interfaces (BCIs) are promising therapies for neurological and psychiatric disorders. However, their broader translation into research and clinical practice remains limited by technological complexity, restricted access to implantable r...

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Low-amplitude intracortical microstimulation produces short-latency modulation throughout the cortical reach-to-grasp network

Marc H Schieber 2026-11-12 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Intracortical microstimulation (ICMS) can be used to deliver information to the cortex from a bidirectional brain-machine interface (BMI), thereby improving control of neuroprosthetic devices. Primary somatosensory cortex (S1) is a common choice for delivering information, but S1-ICMS can...

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Hierarchical causal construction for semantic-granularity-aligned counterfactual medical image generation

Yifan Chen 2026-11-12 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Counterfactual image generation seeks to answer how a medical image should change when a clinical semantic factor is altered, and is important for model interpretation, lesion localization, and controllable image editing. This work addresses the semantic granularity mismatch in existing counterfactu...

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Sub-second dopamine reinforcement shapes juvenile social play

👤 Ying Li 📅 2026-11-12 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Juvenile social play is critical for neurodevelopment, yet the neural mechanisms organizing moment-to-moment interactions remain elusive. Using social behavior sequencing (Social-seq)-an occlusion-robust 3D pose estimation and unsupervised clustering pipeline-we found that nucleus accumbens (NAc) do...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

👤 Hong-Gi Yeom 📅 2026-11-13 📖 1 min 📝 67 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Motor imagery (MI) can support self-initiated control, whereas brain-computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspend...

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A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

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http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

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http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 539 words FMHY

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Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 899 words FMHY

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Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 666 words FMHY

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Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1013 words FMHY

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Graphlink support python now.

/u/graphlinkdev 2026-11-13 1 min 99 words REDDIT PYTHON

Summary: <!-- SC_OFF --> Need to call a GraphQL or REST API and don't want to write the client yourself? Let GraphLink generate it. It's not an AI wrapper, it's a compiler. You give it a schema and it generates a full typed client, with no tokens and no boilerplate. Responses and inp...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xv7202/graphlink_support_python_now/

Why doesn't PyCharm warn about this type mismatch?

/u/Sensitive-Raccoon155 2026-11-13 1 min 119 words REDDIT PYTHON

Summary: <!-- SC_OFF --> 'm using PyCharm with Python 3.13 and trying to implement a generic CQRS pattern. I explicitly specify <code>int</code> as the response type, but PyCharm doesn't report any error when the handler returns <code>str</code>. ``` from abc import ABC, abstr...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xv6qir/why_doesnt_pycharm_warn_about_this_type_mismatch/

The Dawn of the Data Intelligence Engineer

👤 jonbaer 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://motherduck.com/blog/the-data-intelligence-engineer/ Comments URL: https://news.ycombinator.com/item?id=50480480 Points: 81 <...

🔗 **Read full article:**
https://motherduck.com/blog/the-data-intelligence-engineer/

Elevated errors on claude.ai, Claude Code, and platform.claude.com

👤 __vivek 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://status.claude.com/incidents/nn11wk485zfn Comments URL: https://news.ycombinator.com/item?id=50481187 Points: 350 # Comments: 306

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https://status.claude.com/incidents/nn11wk485zfn

AI Is Destroying the Trust Between Writers and Readers

👤 FinnLobsien 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theatlantic.com/books/2026/11/ai-literary-scams-amazon/689391/ Comments URL: <a href="https://news.ycombinator.com/item?id=50481870">https://news.ycombinator.com/item?id=504818...

🔗 **Read full article:**
https://www.theatlantic.com/books/2026/11/ai-literary-scams-amazon/689391/

Doomsday: Friday, 13 November, A.D. 2026 (1960)

👤 voxleone 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.science.org/doi/10.1126/science.132.3436.1291 Comments URL: https://news.ycombinator.com/item?id=50482379 Points: 226 #...

🔗 **Read full article:**
https://www.science.org/doi/10.1126/science.132.3436.1291

Show HN: Reflota – Open-source data recovery for NTFS/FAT disks with phased copy

👤 christ_dev 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/Christ10-8/reflota Comments URL: https://news.ycombinator.com/item?id=50482455 Points: 11 # Comments: 0

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https://github.com/Christ10-8/reflota

Eli Lilly CEO pushes back on Dario Amodei's ludicrous medical claims

👤 Noaidi 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://garymarcus.substack.com/p/breaking-eli-lilly-ceo-becomes-latest Comments URL: https://news.ycombinator.com/item?id=50483480...

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https://garymarcus.substack.com/p/breaking-eli-lilly-ceo-becomes-latest

US test reactor cleared for operations

👤 PaulHoule 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.world-nuclear-news.org/articles/us-test-reactor-cleared-for-operations Comments URL: <a href="https://news.ycombinator.com/item?id=50483533">https://news.ycombinator.co...

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https://www.world-nuclear-news.org/articles/us-test-reactor-cleared-for-operations

Why Zig over C for Systems Programming

👤 torutofu 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.mtsoukalos.eu/2026/11/why-zig-over-c-for-systems-programming/ Comments URL: <a href="https://news.ycombinator.com/item?id=50483611">https://news.ycombinator.com/item?id=50483611...

🔗 **Read full article:**
https://www.mtsoukalos.eu/2026/11/why-zig-over-c-for-systems-programming/

Mercedes-Benz to halt 'all future investments' in Germany

👤 muse900 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.ft.com/content/f4a1fd87-9abc-4965-963a-a5349dbfdab7 Comments URL: https://news.ycombinator.com/item?id=50483875 Points: ...

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https://www.ft.com/content/f4a1fd87-9abc-4965-963a-a5349dbfdab7

The 42-Year Journey to Bring 'Neuromancer' to the Screen

👤 Brajeshwar 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.wired.com/story/exclusive-apple-tv-neuromancer-creator-interviews/ Comments URL: <a href="https://news.ycombinator.com/item?id=50484006">https://news.ycombinator.com/item?i...

🔗 **Read full article:**
https://www.wired.com/story/exclusive-apple-tv-neuromancer-creator-interviews/

Automatic computational reproducibility checks with Metacheck

noreply@blogger.com (Daniel Lakens)
2026-10-30 19 min 3854 words

TWENTY PERCENT STATISTICIAN

Summary: When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

Steve Luck
2024-03-16 1 min 106 words

ERP BOOT CAMP

Summary: The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** <p class="">Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Robot-Mediated Physical Human–Human Interaction in Rehabilitation: A Position Paper

📅 2025-11-25 📖 1 min 📝 187 words REVIEWS BIOMEDICAL ENGINEERING

> **Summary:** Neurorehabilitation conventionally relies on the interaction between a patient and a physical therapist. Robotic systems can improve and enrich the physical feedback provided to patients after neurological injury, but they under-utilize the adaptability and clinical expertise of trained therapists. ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11267500

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

> **Summary:** The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Dynamic muscle proprioception impairment in adolescent idiopathic scoliosis revealed by tendon vibration

📖 1 min 📝 29 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Daniela M Pinzon-Corredor, Christophe Boulay, Alexis Ulian, Guillaume Authier, Nicole Levy, Fabienne Levy, Jean-Luc Jouve, Corentin Petitpas, Sébastien Pesenti, Anne Kavounoudias

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007165?dgcid=rss_sd_all

Altered stability of dynamic functional architecture in the co-occurrence of autism spectrum disorder and attention–deficit/hyperactivity disorder

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Shuting Zhang, Chao Han, Yuning Si, Jinsha Tan, Xin Pan, Kaixin Li, Mengzhu Wang, Xipeng Yue, Xuan Yu, Wei Wei, Yan Bai, Meiyun Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007049?dgcid=rss_sd_all

Do autistic traits modulate the neural processing of logical rule violations in visual and auditory modalities?

📖 1 min 📝 28 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Junyu Wu, Rong Huang, Jingyang Yu, Jiayi Shi, Tianze Wang, Yuting Zhang, Yutong Chen, Jie Li, Shubo Shui, Jing Meng

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006427?dgcid=rss_sd_all

Training interbrain synchrony through dyadic neurofeedback modulates empathy-for-pain

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Mario Francis, Simone Shamay-Tsoory

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006415?dgcid=rss_sd_all

Short-video experience and brain-state dynamics during the processing of rapidly changing video information in youth

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Bijie Tie, Li Geng, Pengcheng Wang, Jason M. Nagata, Jinbo He, Yueyang Xiao, Xueyang Wang, Jiang Qiu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600649X?dgcid=rss_sd_all

Developmental effects of unilateral deafness are ear-independent and persist despite longstanding cochlear implant use

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76151-2

2-AG as a potential biomarker for predicting response to short-term fluoxetine treatment in a mouse model of depression

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41398-026-04548-8

The posterior paraventricular nucleus of the thalamus to nucleus accumbens medial shell circuit mediates an anti-pruritic response

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78362-7

Hippocampal somatostatin neurons gate context generalization during fear memory updating

📅 2026-11-13 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11310-9

Single-neuron encoding of rapidly learned visual information reshapes human perception

👤 Gabriel Kreiman 📅 2026-11-04 📖 1 min 📝 38 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02496-6 The authors show that rapid learning modulates neuronal activity in the human visual cortex, requiring extra processing time to support recog...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02496-6

The human claustrum tracks uncertainty and prediction error

📅 2026-11-04 📖 1 min 📝 54 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02471-1 Single-neuron recordings provide direct evidence that the human claustrum tracks uncertainty and prediction errors during adaptive behavior. ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02471-1

Information formatting in neuronal populations predicts differential functional contributions to behavior

👤 Marlene R. Cohen 📅 2026-11-10 📖 1 min 📝 32 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 10 November 2026; doi:10.1038/s41593-026-02514-7 Using experiments in monkeys and modeling, the authors demonstrate that the formatting of population information predicts the contribution of...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02514-7

An explainable deep learning framework for differentiating idiopathic normal pressure hydrocephalus from Alzheimer's disease using brain MRI

Sofyan Freihat 2026-11-10 1 min 269 words FRONTIERS NEUROINFORMATICS

Summary: Background and objectivesIdiopathic normal pressure hydrocephalus (NPH) and Alzheimer's disease (AD) may present with overlapping clinical and neuroimaging features, complicating differential diagnosis. This study evaluated an explainable, attention-enhanced deep learning framework for distinguishin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1990170

Perceived sound quality and speech recognition are not interchangeable outcome measures in adult cochlear implant users

Andreas Büchner 2026-11-10 1 min 198 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis study examined whether descriptor-based reports of perceived sound quality in adult cochlear implant users are associated with speech-recognition outcomes, preoperative maximum monosyllabic word recognition, and patient- or device-related factors.MethodsIn this retrospective single-...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1968712

Assessing systematic spatial differences between intraoperative and postoperative Probabilistic Sweet Spots in Deep Brain Stimulation for Essential Tremor

Simone Hemm 2026-11-10 1 min 245 words FRONTIERS HUMAN NEUROSCIENCE

Summary: InroductionDeep Brain Stimulation (DBS) is an established therapy for movement disorders such as Essential Tremor (ET). While electrode displacement across intraoperative and postoperative stages has been studied, Probabilistic Sweet Spot (PSS) analyses rarely account for temporal evolution. Consequ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2005438

Effects of intermittent theta burst stimulation combined with hyperbaric oxygen therapy on cognitive function and cerebral blood flow in patients with post-stroke mild cognitive impairment

Min Tang 2026-11-10 1 min 371 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo observe the effects of intermittent theta burst stimulation (iTBS) combined with hyperbaric oxygen therapy on cognitive function and cortical hemodynamics in patients with post-stroke mild cognitive impairment using functional near-infrared spectroscopy (fNIRS).MethodsSixty patients with...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966062

Neural architecture of the jhānas: a gamma-infraslow duet with quantum overtones

Paul Dennison

2026-11-13 | 1 min | 331 words | FRONTIERS HUMAN NEUROSCIENCE

Summary: Building on previous electrophysiological investigations that posited meditative absorption (jhāna) as a radical disruption of sensory default consciousness (DCs), this study explores the underlying non-linear neural architectures and state transitions. Re-analysis of long-duration (n = 18) DC-stabi...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966888

Source-aware forward modelling and detection limits of acoustoelectric neural recording

Christopher Chare, Rachid Haouari, Klara Volckaert, Agknaton Bottenberg, Sebastian Haesler, Nick Van Helleputte, Xavier Rottenberg, Samer Houri and Peter Peumans

2026-11-10 | 1 min | 298 words | JOURNAL NEURAL ENGINEERING

Summary: Objective. Acoustoelectric recording uses focused ultrasound to select the region sampled by scalp or implanted electrodes. We relate its recorded voltage to source structure, terminal conditions and the carrier phase of the focus, and identify configurations that favour neural recording. Approach. ...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aebb68

Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

E Canario, C Shearer, M Akcakaya, D Weber, S Chase and J L Collinger

2026-11-10 1 min

291 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally oc...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aebb65

Closed-loop neuroprosthetic control through spared neural activity enables proportional foot movements after spinal cord injury

👤 Vlad Cnejevici, Matthias Ponfick, Dietmar Fey, Raul C Sîmpetru and Alessandro Del Vecchio

📅 2026-11-10 📖 1 min 📝 277 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Loss of voluntary foot movement after spinal cord injury (SCI) can significantly limit independent mobility and quality of life. To improve motor output after injury, functional electrical stimulation (FES) is used to deliver stimulation pulses through the skin to affected muscles. While ...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aebb64

Golgi: an open-source graphical platform for image-to-recruitment modeling of peripheral nerve stimulation

David Lung, Yuting Jia, Roland Blumer, Lukas Reissig, Lydia M Zopf, Patrick Heimel, Christoph Kraus, Andrea Moro, Matteo Fachino and Max Haberbusch

2026-11-10 1 min 289 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Computational models of peripheral nerve stimulation—coupling finite-element bioelectric fields to biophysical axon models—are essential for designing electrodes and waveforms for neuromodulation, but established open tools are code-first, assume substantial modeling expertise, and often ...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aeba17

Recent advances and challenges in autonomous wireless implantable neurologgers for animal studies: neural recording, power supply, and data transmission technologies

👤 Ye Huang, Jiaxu Cao, Xuanqi Wang, Xiaoli You, Ruiyu Bai, Mingyang Li, Yuxiang Hong, Honglong Chang, Haitao Zhao, Minghao Wang, Hao Zhang, Jun Chen and Bowen Ji

📅 2026-11-10 📖 1 min 📝 136 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Wireless implantable neurologgers are emerging as miniature autonomous systems for long-term, untethered neural recording in animal studies. However, their development remains constrained by critical engineering bottlenecks, including neural recording, power supply, and data transmission....

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb97d

Augmented reality paradigm and EA-DGCNN decoding algorithm for motor imagery in stroke patients

Fangzhou Xu, Di Yan, Chenggong Zhu, Meiling Yu, Shi Huang, Yunhuan Wang, Runfa Liu, Kaixiang Hu, Jiancai Leng, Yang Zhang, Chao Feng and Tzyy-Ping Jung

2026-11-13 1 min 252 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Stroke represents one of the leading causes of long-term motor dysfunction on a global scale. Existing motor imagery (MI) brain–computer interface (BCI) rehabilitation training systems face critical challenges of suboptimal patient engagement and limited electroencephalography (EEG) featu...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aebdce

A phase-calibrated combined electrical-sensory neuromodulation platform for short-term frontoparietal network modulation and cognitive fatigue resistance

👤 Chenfei Yan, Yifan Ouyang, Shuo Zhang, Bin Jing, Zhongqi Liu and Qianxiang Zhou

📅 2026-11-13

📖 1 min 📝 192 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Non-invasive combined electrical-sensory neuromodulation may be limited by cross-modal timing discrepancies arising from modality-specific conduction delays. This pilot study aimed to develop and evaluate an individualized delay-calibrated stimulation platform integrating transcranial alt...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aebcb6

DSGTANet: a semi-supervised dual-stream graph-temporal adaptation network for cross-subject continuous EEG emotion recognition

Chen Lu, Li-Lin Jie, Ying Xiong, Jia Yuan and Wang-yang Xu

2026-11-13 1 min 316 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Continuous emotion recognition using electroencephalogram (EEG) can be formulated as valence and arousal regression in a two-dimensional affective space rather than discrete emotion classification. It has applications in brain–computer interfaces (BCIs), including mental-disorder regulati...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeba18

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113012455&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113012455&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113012455&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113012455&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113012455&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261113012455&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Influencing Factors of Positional Stability of Implantable Collamer Lenses in Terms of Rotation, Decentration, and Tilt in Patients With Myopia

👤 Jing Zhao 📅 2026-11-12 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSIONS: ICL positional stability was within an acceptable range. ICL fixation angle and the structures of the iris and ciliary body influenced the rotation, decentration, and tilt of the ICL. Abnormal vault values were also correlated with ICL rotation and tilt.

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Visual Outcomes of Autologous RPE-Based Transplantation for Massive Submacular Hemorrhage in Neovascular AMD and PCV: A Systematic Review and Meta-Analysis

👤 Huijin Chen
📅 2026-11-12 📖 1 min 📝 48 words

LOW VISION

Summary: CONCLUSIONS: Autologous RPE-based transplantation may offer salvage value for highly selected eyes with AMD/PCV-related SMH. However, the primary prediction interval crossed no mean change, sensitivity results depended on follow-up definition and cohort inclusion, and complications were frequent. He...

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Effects of emotion and emotional congruence on the multisensory temporal binding window during audiovisual speech integration

Ryan A Stevenson 2026-11-12 1 min 69 words LOW VISION

Summary: The integration of auditory and visual cues supports efficient speech and emotion perception, yet little is known about how emotional salience interacts with temporal congruency during audiovisual binding. In this study, 125 participants completed a synchrony judgment task using dynamic face-voice p...

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Seeing Text Clearly in the Dark: Unifying Localization and Enhancement via Deep Morphological Synergy

Rushi Lan 2026-11-12 1 min 67 words LOW VISION

Summary: Low-light text localization, an interdisciplinary field, has gained significant attention recently. The goal is to provide enhanced text locational information beneficial for both machine and human vision. Existing one-stage and two-stage methods either rely on single detectors specially designed fo...

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RPDL-YOLO: An efficient YOLOv8 framework for small bird detection in airport airside environments

👤 Heng Yang 📅 2026-11-12 📖 1 min 📝 63 words LOW VISION

Summary: Bird intrusions into airport security zones pose significant threats to aviation safety. However, the small size, rapid movement, and irregular flight patterns of birds often result in missed detections and false alarms in vision-based surveillance systems, especially in complex open-sky environment...

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Mobile Health Apps for Individuals Who Are Blind or Have Low Vision: Systematic Review of Evidence, Accessibility, and App Quality

👤 Jordi Miró 📅 2026-11-12 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: The current landscape of mobile health apps for individuals who are blind or have low vision remains limited, uneven in accessibility and quality, and supported by a limited and heterogeneous evidence base. The absence of identified apps targeting psychological functioning represents a ...

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DPCoOp: Dual positive context optimization for multi-label recognition with limited annotations

👤 Yingchun Kuang 📅 2026-11-12 📖 1 min 📝 67 words LOW VISION

Summary: Multi-label recognition with limited annotations has attracted significant attention due to its close alignment with real-world needs. Vision-language pretraining models like CLIP have effectively mitigated inadequate feature learning from scarce annotations, making them a central research focus in ...

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Automated detection of blink reflexes evoked by optogenetic stimulation of TRPV1-lineage corneal afferents in transgenic mice

👤 Carl Y Saab 📅 2026-11-12 📖 1 min 📝 28 words LOW VISION

Summary: CONCLUSIONS: This video-based method enables accurate, high-throughput, and observer-independent empirical assessment of stimulus-evoked blink reflexes, providing a quantitative platform for investigating corneal sensory responses relevant to ocular nociception.

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Step change in glaucoma polygenic risk score performance enables clinical utility

Jamie E Craig 2026-11-12 1 min 58 words LOW VISION

Summary: CONCLUSIONS: This enhanced PRS, available for clinical use through an accredited genotyping workflow, enables effective risk stratification for OAG across diverse ancestries, facilitating earlier identification and treatment of high-risk individuals while supporting resource rationalisation for thos...

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A robust framework for cross-level spatial-frequency collaborative recognition in plant disease images

Haiming Lan 2026-11-12 1 min 61 words LOW VISION

Summary: Plant disease images commonly suffer from pixel confusion, texture blurring and semantic ambiguity caused by cluttered backgrounds, noise interference and diverse lesion morphologies. These intra- and inter-class disturbances break the representation balance between global context and local details,...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

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Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

👤 Lei Huang 📅 2026-11-11 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

👤 Sevda Uzun 📅 2026-11-12 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Prefrontal transcranial direct current stimulation combined with cognitive training reduces anxiety, daytime sleepiness, and cognitive complaints in women after COVID-19: A randomized, double-blind, sham-controlled trial

👤 Guilherme Augusto Santos Bueno
📅 2026-11-12 📖 1 min 📝 37 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Adding left-prefrontal tDCS to cognitive training reduced anxiety and daytime sleepiness and improved cognition and psychomotor speed in women with post-COVID cognitive complaints, with parallel changes in prefrontal EEG markers. Confirmation in larger samples is needed.

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Home-based transcranial direct current stimulation and exercise for pain in people with knee osteoarthritis: A mixed-methods pilot randomised controlled trial

Rocco Cavaleri 2026-11-12 1 min 66 words TDCS TACS TRNS

Summary: This mixed-methods pilot trial evaluated the feasibility and patient perceptions of home-based transcranial direct current stimulation (tDCS) combined with exercise for pain in people with knee osteoarthritis. Twenty-four participants were allocated to active or sham home-based tDCS plus exercise. P...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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Acute aerobic exercise and orbitofrontal cortex reactivity during positive emotion processing in obesity: An exploratory fNIRS study

👤 Jun Yan 📅 2026-11-12 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSION: Obese students showed lower baseline PA and right OFC activation during active positive elaboration; post-exercise PA and right OFC activation were higher in the same context. These single-arm associations cannot be attributed to exercise; right OFC is a preliminary trial signal, dlPFC i...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

👤 Jilai Li 📅 2026-11-11 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Spinal cord-computer interfaces for neurorehabilitation: Technical challenges and clinical perspectives

Minfei Wu 2026-11-11 1 min 35 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: This Perspective aims to clarify the clinical and engineering scope of spinal cord-computer interfaces (SCCIs) in neurorehabilitation, distinguishing them from adjacent brain-computer interface, neuromodulation, and assistive-technology approaches while identifying the key barriers to tra...

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Clinical safety assessment of implanted brain-computer interfaces: current evidence and a pathway to standardization

Wen-Jun Tu 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Implanted brain-computer interfaces are beginning to move beyond short laboratory sessions into long-term and home use. The evidence used to judge clinical safety remains fragmented. This review examines penetrating intracortical, cortical-surface (including subpial) and cerebral-venous systems, wit...

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Characteristics of EEG Microstate Analysis of Power Grip and Precision Grip Hand Movements from Motor Planning to Execution

👤 Gang Wang
📅 2026-11-12 📖 1 min 📝 63 words

BRAIN COMPUTER INTERFACE

Summary: Hand movement is a fundamental daily motor behavior. Precision grip and power grip reflect distinct post-stroke recovery stages and neuroplasticity, making their neural mechanisms critical for motor control and neurorehabilitation. This study employed electroencephalographic (EEG) microstate analysi...

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Robotic Gait Multi-channel Mu Beta Neurofeedback Training in Paraplegics Enhanced Inter-regional Connectivity and Sensorimotor Functions

👤 Denis Delisle-Rodriguez
📅 2026-11-12 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Complete spinal cord injury (SCI) that severely affects somatosensory and motor functions, greatly challenges advanced neuro-technologies and therapies addressed to repair neural system pathways and recover lost functions. Non-invasive solutions have been explored, reaching encouraging results. This...

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A translational platform for brain-computer interfaces and adaptive neuromodulation: Technical characterization, long-term validation, and implementation of the CorTec brain interchange-BCI2000 ecosystem

👤 Kai J Miller
📅 2026-11-12 📖 1 min 📝 59 words

BRAIN COMPUTER INTERFACE

Summary: Adaptive neuromodulation systems and implantable brain-computer interfaces (BCIs) are promising therapies for neurological and psychiatric disorders. However, their broader translation into research and clinical practice remains limited by technological complexity, restricted access to implantable r...

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The potential of transcranial direct current stimulation to enhance P300-based brain-computer interface speller performance: A feasibility study

👤 Gabriel Pires
📅 2026-11-12 📖 1 min 📝 33 words

BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: To evaluate whether anodal transcranial direct current stimulation (tDCS) over the left dorsolateral prefrontal cortex (DLPFC) enhances the P300 event-related potential (ERP) and whether this enhancement translates into improved target-non-target automatic classification.

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Low-amplitude intracortical microstimulation produces short-latency modulation throughout the cortical reach-to-grasp network

Marc H Schieber 2026-11-12 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Intracortical microstimulation (ICMS) can be used to deliver information to the cortex from a bidirectional brain-machine interface (BMI), thereby improving control of neuroprosthetic devices. Primary somatosensory cortex (S1) is a common choice for delivering information, but S1-ICMS can...

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Hierarchical causal construction for semantic-granularity-aligned counterfactual medical image generation

Yifan Chen 2026-11-12 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Counterfactual image generation seeks to answer how a medical image should change when a clinical semantic factor is altered, and is important for model interpretation, lesion localization, and controllable image editing. This work addresses the semantic granularity mismatch in existing counterfactu...

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Sub-second dopamine reinforcement shapes juvenile social play

Ying Li 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Juvenile social play is critical for neurodevelopment, yet the neural mechanisms organizing moment-to-moment interactions remain elusive. Using social behavior sequencing (Social-seq)-an occlusion-robust 3D pose estimation and unsupervised clustering pipeline-we found that nucleus accumbens (NAc) do...

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Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

2026-05-18 1 min 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000346

Friday Daily Thread: r/Python Meta and Free-Talk Fridays

/u/ AutoModerator | 2026-11-13 | 1 min | 212 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Meta Discussions and Free Talk Friday 🎙️</h1> Welcome to Free Talk Friday on /r/Python! This is the place to discuss the r/Python community (meta discussions), Python news, projects, or anything else...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xusw5u/friday_daily_thread_rpython_meta_and_freetalk/

For devs using Claude Code or similar at work: what's the 10% you still fix by hand in Python?

/u/ Quantititoo | 2026-11-13 | 1 min | 111 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I keep hearing that AI now writes most of the code at some companies. For people actually shipping Python with Claude Code, Cursor or Copilot: <ol> Roughly what share of your merged code starts as AI output? What do you most often have to correct? B...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xusxv9/for_devs_using_claude_code_or_similar_at_work/

Show HN: RealtimeVAD – trading false positives for 110ms earlier end-of-speech

👤 koljab 📅 2026-11-12 📖 1 min 📝 181 words HACKER NEWS

> **Summary:** Hey guys. I recently watched the latency reports for my local voice assistant, which show a waveform plus silero probability for voice activity. I noticed silero lags visibly on the speech off signal. There had to be a method to get this signal closer to what I felt is optimal. My idea was to u...

🔗 **Read full article:**
https://github.com/KoljaB/RealtimeVAD

Edge0-145B: Frontier AI Capabilities on Your Own Computer

👤 sroussey 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://edge0.ai/blog/introducing-edge0-145b-frontier-ai-capabilities-on-your-own-computer Comments URL: <a href="https://news.ycombinator.com/item?id=50475713">https://new...

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https://edge0.ai/blog/introducing-edge0-145b-frontier-ai-capabilities-on-your-own-computer

Nvidia employees petition to bring back Jensen Huang

👤 lavendermon 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.wired.com/story/nvidia-employees-petition-for-jensen-huangs-return/ Comments URL: <a href="https://news.ycombinator.com/item?id=50477231">https://news.ycombinator.com/item...

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https://www.wired.com/story/nvidia-employees-petition-for-jensen-huangs-return/

Brave Search API now accepts agentic payments via the x402 protocol

👤 sjdh76 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://brave.com/blog/search-api-agentic-payments/ Comments URL: https://news.ycombinator.com/item?id=50477649 Points: 35 # Comments: 11...

🔗 **Read full article:**
https://brave.com/blog/search-api-agentic-payments/

Switcher (2004)

👤 sammatilda 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://folklore.org/Switcher.html Comments URL: https://news.ycombinator.com/item?id=50477714 Points: 71 # Comments: 15

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https://folklore.org/Switcher.html

IceWM

👤 inatreecrown2 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ice-wm.org/ Comments URL: https://news.ycombinator.com/item?id=50478515 Points: 41 # Comments: 19

🔗 **Read full article:**
https://ice-wm.org/

Show HN: PharmA2A, an open protocol for pharmacy-to-supplier orders on A2A

👤 eliefo 📅 2026-11-13 📖 1 min 📝 84 words HACKER NEWS

> **Summary:** I'm a pharmacist in France and a nerd. I built an open source protocol based on A2A, Called PharmA2A. Why, because normally I order directly from more than hundred of suppliers, by phone mails or on the web forms. The idea is to automatise orders of selling and buying between pharmacies and thei...

🔗 **Read full article:**
https://github.com/pharma2a/pharma2a

The average cost for a new vehicle in the U.S. hits a record high: over $51,000

👤 blinding-streak 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.npr.org/2026/11/10/nx-s1-6026659/average-vehicle-price-record-high Comments URL: <a href="https://news.ycombinator.com/item?id=50478622">https://news.ycombinator.com/item?i...

🔗 **Read full article:**
https://www.npr.org/2026/11/10/nx-s1-6026659/average-vehicle-price-record-high

Scientists claim the universe is twice as old as believed

👤 mattm 📅 2026-11-13 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://geekspin.co/scientists-claim-the-universe-is-twice-as-old-as-believed/ Comments URL: <a href="https://news.ycombinator.com/item?id=50478635">https://news.ycombinator.com/item?i...

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Advancing Embodied Intelligence in Robotic-Assisted Endovascular Procedures: A Systematic Review of AI Solutions

📅 2025-12-16 📖 1 min 📝 191 words REVIEWS BIOMEDICAL ENGINEERING

> **Summary:** Endovascular procedures have revolutionized vascular disease treatment, yet their manual execution is challenged by the demands for high precision, operator fatigue, and radiation exposure. Robotic systems have emerged as transformative solutions to mitigate these inherent limitations. A crucial mom...

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http://ieeexplore.ieee.org/document/11301748

TDP-43 dysfunction destabilizes glial homeostatic identity in ALS/FTD

📖 1 min 📝 14 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Anna E Dale, Rita Sattler

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https://www.sciencedirect.com/science/article/pii/S0006899326004804?dgcid=rss_sd_all

PKR drives neuropathology in a mouse model of Leigh syndrome

👤 Gonzalez-Torres, M., Bianchi, P., Blanco-Ramos, M., Luna-Sanchez, M., Prada-Dacasa, P., Puighermanal, E., Canal, C., Cutando, L., Le, S., Gella, A., Girona, M., Sanchez-Benito, L., Armengol, B., Appiah, I., Fernandez-Irigoyen, J., Alkhatib, M., Feist, L., McElroy, G., Lisowski, P., del Sol, A., Santamaria, E., Prigione, A., Sanz, E., Quintana, A.

📅 2026-11-12 📖 1 min 📝 148 words BIORXIV NEUROSCIENCE

Summary: Mitochondrial dysfunction is increasingly recognized as a trigger of innate immune responses, yet whether these responses contribute to neurodegeneration remains poorly understood. Here, we uncover a cell-type-specific antiviral program in vulnerable GABAergic neurons of Ndufs4-deficient mice and hu...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765677v1?rss=1

Systematic and dynamic task coding in the frontal cortex

Park, J., Park, H.-B., Holmes, C. D., Snyder, L. H.

2026-11-12 1 min 195 words

BIORXIV NEUROSCIENCE

Summary: Behavioral flexibility depends on the capacity to form and utilize task representations. Prior work has shown that baseline activity in the prefrontal cortex (PFC) can differ between two tasks. Comparing more than two tasks makes it possible to ask whether different neuronal populations share a comm...

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The thalamic reticular nucleus provides spatially and temporally tuned inhibition to the lateral geniculate nucleus across species

Miao, Y., Ciftcioglu, U. M., Gorin, A. S., Dakwala, R., Ahn, S., Sommer, F. T., Hirsch, J. A.

2026-11-12 1 min 177 words

BIORXIV NEUROSCIENCE

Summary: By proposing a role in spatial attention, Francis Crick piqued interest in the thalamic reticular nucleus (TRN), a GABAergic structure that receives input from relay cells projecting to cortex and feeds back inhibition in return. Importantly, predictions of Crick's hypothesis extend beyond attention...

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Multi-omics characterization of white matter free water reveals genetic and vascular determinants of neurological disorders

👤 Chang ChengYuqin HeXi CaoShizhen XiangYuanyuan FangMengying WangZhonghe ShaoYao YaoHao HuangShabei XuXiang LuoXiaopeng ZongWei WangXingjie HaoMinghuan Wangahttps://ror.org/00p991c53Department of Neurology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, Chinabhttps://ror.org/00p991c53Hubei Key Laboratory of Neural Injury and Functional Reconstruction, Huazhong University of Science and Technology, Wuhan 430030, Chinachttps://ror.org/00p991c53Department of Epidemiology and Biostatistics, Ministry of Education Key Laboratory of Environment and Health, School of Public Health, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, Chinadhttps://ror.org/030bhh786School of Biomedical Engineering & State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, ChinaeShanghai Clinical Research and Trial Center, Shanghai 201203, China

📅 2026-11-11 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 46, November 2026.
SignificanceWhite matter free water (FW-WM) is a brain imaging measure that can capture changes in extracellular water content and microstructural alterations in white matter. However, its genetic underpinnin...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2611353123?af=R

PPP1R1B differentially modulates alcohol intake and emotional comorbidity through distinct prefrontal circuits

👤 Fei LiMinghui ChenXiang LiZunyue ZhangHuayou LuoMei ZhuHuawei WangYu XuFengrong ChenKunhua Wangahttps://ror.org/00xyeez13Faculty of Life science and Technology, Kunming University of Science and Technology, Kunming 650500, Chinabhttps://ror.org/00xyeez13Medical school, Kunming University of Science and Technology, Kunming 650500, Chinachttps://ror.org/0040axw97Yunnan Technological Innovation Center of Drug Addiction Medicine, Yunnan University, Kunming 650500, Chinadhttps://ror.org/0040axw97School of Medicine, Yunnan University, Kunming 650500, Chinaehttps://ror.org/02g01ht84Department of Gastrointestinal and Hernia Surgery, The First Affiliated Hospital of Kunming Medical University, Kunming 650032, Chinafhttps://ror.org/02g01ht84Department of Ultrasound, The First Affiliated Hospital of Kunming Medical University, Kunming 650032, Chinaghttps://ror.org/02g01ht84Reproductive Genetics Department, The First Affiliated Hospital of Kunming Medical University, Kunming 650032, China

📅 2026-11-11 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 46, November 2026.
SignificanceExcessive alcohol consumption frequently co-occurs with anxiety and depression in alcohol use disorder (AUD), yet the specific neural circuits and molecular mechanisms coordinating these comorbid ...

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https://www.pnas.org/doi/abs/10.1073/pnas.2611266123?af=R

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

👤 Ansel Y. Rodríguez-González
📅 2026-10-07 📖 1 min 📝 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1919962

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

👤 Tingwei Quan
📅 2026-10-07 📖 1 min 📝 139 words

FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

👤 Bocheng Wang 📅 2026-10-07 📖 1 min 📝 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1869687

Editorial: Exploring the role of criticality in neural systems

👤 Tiago L. Ribeiro 📅 2026-11-04 📖 1 min 📝 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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Multi-receptor whole-brain modeling reveals affinity-driven chemoarchitectural mechanism of virtual atomoxetine action

👤 Gustavo Deco
📅 2026-11-04 📖 1 min 📝 219 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Linking molecular mechanisms with large-scale brain dynamics remains a fundamental challenge in computational psychiatry. Receptor-informed whole-brain models have provided mechanistic insights into pharmacological interventions, yet they have largely relied on single-receptor topographies that cann...

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MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

👤 Haoyan Liu
📅 2026-10-30 📖 1 min 📝 229 words

FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

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MA-Pose and DMMS-GCN: skeleton-based martial-arts category recognition with edge-device deployment

Shilin Guo 2026-11-04 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: With the development of visual sensor technology and human pose estimation algorithms, skeleton-based action recognition has received extensive attention in sports analysis and training systems. The recognition of martial-arts actions is difficult because different styles can share similar poses whi...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1967140

Deep reinforcement learning–driven model-based adaptive PID strategy for six-wheeled rocker–bogie mobile platform

Hongxue Feng 2026-11-04 1 min 202 words FRONTIERS NEUROROBOTICS

Summary: IntroductionSix-wheel rocker–bogie rovers encounter nonlinear dynamics, variable traction, and wheel slip that can degrade the performance of fixed-gain proportional–integral–derivative (PID) controllers. This study develops a hybrid deep reinforcement learning–PID framework to improve steering and ...

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Reliability-aware multimodal sensor fusion for human motion monitoring with safety-constrained neurorobotic feedback in sports training

Jiaoyu Wang 2026-11-04 1 min 247 words FRONTIERS NEUROROBOTICS

Summary: IntroductionEmbodied intelligent systems require reliable perception and safety-constrained feedback mechanisms for human motion understanding under heterogeneous sensing conditions. In this study, volleyball training is considered as a representative dynamic human-motion scenario rather than the so...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Comparison of outcome in patients with idiopathic intracranial hypertension (IIH) following optic nerve sheath fenestration (ONSF) using a medial transconjunctival approach with dural window or superomedial lid-crease incision approach

👤 Philip A Tenzel
📅 2026-11-12 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: This study demonstrates that the superomedial lid-crease approach with fenestration and the medial transconjunctival approach with dural window for ONSF are both effective in preventing vision loss in patients with IIH, with similarly low side-effect profiles.

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Geographic concordance of cataract and glaucoma burden and associated country-level exposures across 204 countries and territories

Changzheng Chen 2026-11-12 1 min 60 words LOW VISION

Summary: CONCLUSION: National cataract- and glaucoma-related vision-loss burdens showed geographic concordance. The exposure findings were country-level associations, and their statistical support varied across regional and spatial models. These geographic patterns may help inform planning for cataract and g...

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Dynamic receptive field and feature enhanced network for bridge defect detection

Hui Li 2026-11-12 1 min 50 words LOW VISION

Summary: INTRODUCTION: Accurate defect detection in underwater and cross-domain marine infrastructure, such as bridge piers, submarine pipelines, and subsea equipment, is essential for structural safety assessment and intelligent ocean engineering. Existing methods, however, suffer from large scale variation...

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Demographic and geographic variations in US blindness and low vision mortality, 1999-2020

👤 Aalaa Saleh 📅 2026-11-12 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSION: Blindness- and low-vision-related mortality in the United States exhibit strong age dependence and persistent demographic and geographic disparities. The recent stagnation in the decline in mortality highlights the need for targeted, equity-focused public health interventions to reduce p...

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Influencing Factors of Positional Stability of Implantable Collamer Lenses in Terms of Rotation, Decentration, and Tilt in Patients With Myopia

👤 Jing Zhao 📅 2026-11-12 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSIONS: ICL positional stability was within an acceptable range. ICL fixation angle and the structures of the iris and ciliary body influenced the rotation, decentration, and tilt of the ICL. Abnormal vault values were also correlated with ICL rotation and tilt.

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Visual Outcomes of Autologous RPE-Based Transplantation for Massive Submacular Hemorrhage in Neovascular AMD and PCV: A Systematic Review and Meta-Analysis

👤 Huijin Chen
📅 2026-11-12 📖 1 min 📝 48 words LOW VISION

Summary: CONCLUSIONS: Autologous RPE-based transplantation may offer salvage value for highly selected eyes with AMD/PCV-related SMH. However, the primary prediction interval crossed no mean change, sensitivity results depended on follow-up definition and cohort inclusion, and complications were frequent. He...

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Effects of emotion and emotional congruence on the multisensory temporal binding window during audiovisual speech integration

Ryan A Stevenson 2026-11-12 1 min 69 words LOW VISION

Summary: The integration of auditory and visual cues supports efficient speech and emotion perception, yet little is known about how emotional salience interacts with temporal congruency during audiovisual binding. In this study, 125 participants completed a synchrony judgment task using dynamic face-voice p...

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Seeing Text Clearly in the Dark: Unifying Localization and Enhancement via Deep Morphological Synergy

Rushi Lan 2026-11-12 1 min 67 words LOW VISION

Summary: Low-light text localization, an interdisciplinary field, has gained significant attention recently. The goal is to provide enhanced text locational information beneficial for both machine and human vision. Existing one-stage and two-stage methods either rely on single detectors specially designed fo...

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RPDL-YOLO: An efficient YOLOv8 framework for small bird detection in airport airside environments

👤 Heng Yang 📅 2026-11-12 📖 1 min 📝 63 words LOW VISION

Summary: Bird intrusions into airport security zones pose significant threats to aviation safety. However, the small size, rapid movement, and irregular flight patterns of birds often result in missed detections and false alarms in vision-based surveillance systems, especially in complex open-sky environment...

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Mobile Health Apps for Individuals Who Are Blind or Have Low Vision: Systematic Review of Evidence, Accessibility, and App Quality

👤 Jordi Miró 📅 2026-11-12 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: The current landscape of mobile health apps for individuals who are blind or have low vision remains limited, uneven in accessibility and quality, and supported by a limited and heterogeneous evidence base. The absence of identified apps targeting psychological functioning represents a ...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

Paul M Briley 2026-11-10 1 min 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

👤 Hamed Ekhtiari 📅 2026-11-10 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

👤 Shouping Zhao 📅 2026-11-11 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

👤 Lei Huang 📅 2026-11-11 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

👤 Zahra Hadi Alqallaf 📅 2026-11-11 📖 1 min 📝 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

👤 Kanecia O Zimmerman 📅 2026-11-11 📖 1 min 📝 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

Sevda Uzun 2026-11-12 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou
2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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https://pubmed.ncbi.nlm.nih.gov/43013564/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261112190000&v=2.20.1

Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

Junhua Li 2026-11-10 1 min 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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https://pubmed.ncbi.nlm.nih.gov/43008527/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261112185957&v=2.20.1

Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger 2026-11-11 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this animal, condition-dependent dimensionality was significantly higher in F5 than in AIP or M1, while alignment and reconstruction analyses showed descriptive area-specific patterns in how closely combined-finger activity resembled a simple combination of single-finger activity. For...

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Effects of brain-computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

👤 Lili Fang 📅 2026-11-11 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Brain-computer interface-triggered functional electrical stimulation showed a clearer signal of benefit vs. conventional rehabilitation than brain-computer interface-controlled robotic. However, the comparison between the two active modalities relies entirely on indirect evidence from a ...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

👤 Jilai Li 📅 2026-11-11 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Spinal cord-computer interfaces for neurorehabilitation: Technical challenges and clinical perspectives

Minfei Wu 2026-11-11 1 min 35 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: This Perspective aims to clarify the clinical and engineering scope of spinal cord-computer interfaces (SCCIs) in neurorehabilitation, distinguishing them from adjacent brain-computer interface, neuromodulation, and assistive-technology approaches while identifying the key barriers to tra...

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Clinical safety assessment of implanted brain-computer interfaces: current evidence and a pathway to standardization

Wen-Jun Tu 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Implanted brain-computer interfaces are beginning to move beyond short laboratory sessions into long-term and home use. The evidence used to judge clinical safety remains fragmented. This review examines penetrating intracortical, cortical-surface (including subpial) and cerebral-venous systems, wit...

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Characteristics of EEG Microstate Analysis of Power Grip and Precision Grip Hand Movements from Motor Planning to Execution

👤 Gang Wang
📅 2026-11-12 📖 1 min 📝 63 words

BRAIN COMPUTER INTERFACE

Summary: Hand movement is a fundamental daily motor behavior. Precision grip and power grip reflect distinct post-stroke recovery stages and neuroplasticity, making their neural mechanisms critical for motor control and neurorehabilitation. This study employed electroencephalographic (EEG) microstate analysi...

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Robotic Gait Multi-channel Mu Beta Neurofeedback Training in Paraplegics Enhanced Inter-regional Connectivity and Sensorimotor Functions

Denis Delisle-Rodriguez 2026-11-12 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Complete spinal cord injury (SCI) that severely affects somatosensory and motor functions, greatly challenges advanced neuro-technologies and therapies addressed to repair neural system pathways and recover lost functions. Non-invasive solutions have been explored, reaching encouraging results. This...

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https://pubmed.ncbi.nlm.nih.gov/43019280/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261112185957&v=2.20.1

Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

22-Year-Old PostgreSQL Bug Turned into Remote Code Execution

👤 ozirus 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://mehmetince.net/part-3-6-how-i-ended-up-turning-a-22-year-old-postgresql-type-confusion-primitive-into-rce-cve-2026-6460/ Comme...

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Rust 1.100

👤 mbStavola 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.rust-lang.org/2026/11/12/Rust-1.100.0/ Comments URL: https://news.ycombinator.com/item?id=50472038 Points: 320 # Comments: 1...

🔗 **Read full article:**
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Gemini 4 Hydrogen and Orchestrator

👤 jonbaer 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://antigravity.google/blog/gemini-4-hydrogen-in-google-antigravity/ Comments URL: <a href="https://news.ycombinator.com/item?id=50472260">https://news.ycombinator.com/item?id=50472260</...

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https://antigravity.google/blog/gemini-4-hydrogen-in-google-antigravity/

Dwarf planet candidate appears which takes roughly 280 years to orbit our sun

👤 cybermango 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** <p>Article URL: <a href="https://www.smithsonianmag.com/smart-news/astronomers-found-a-mysterious-object-orbiting-far-beyond-pluto-it-could-be-a-new-dwarf-planet-hidden-in-our-solar-system-180989857/">https://www.smithsonianmag.com/smart-news/astronomers-found-a-mysterious-object-orbiting-far-beyond...

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Distributed Ray Tracer

👤 juliusgeo 📅 2026-11-12 📖 1 min 📝 192 words HACKER NEWS

> **Summary:** <p>Hey HN, I'm Julius, I'm a backend engineer at Hatchet and created this demo to show off some of Hatchet's new features, primarily durable streams and serverless endpoints. Those, however, are just background details that made this significantly easier to scale. By distributing the rendering acros...

🔗 **Read full article:**
https://ray.hatchet.run

Show HN: Cerulion – ROS 2-compatible robotics runtime with multi-process replay

👤 lakshays 📅 2026-11-12 📖 3 min 📝 702 words HACKER NEWS

> **Summary:** Hey HN! I'm Lakshay. Raveen and I have built Cerulion, a ROS2-compatible robotics runtime written in Rust. We've worked hard on lowering IPC transport latencies without making the single process tradeoff or forcing POD message types. These fast multi-process graphs can then be resimmed determinis...

🔗 **Read full article:**
https://github.com/cerulion-inc/cerulion

AI is turning Apple's app review process into a nightmare for synth developers

👤 anigbrowl 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://synthanatomy.com/2026/11/ai-is-turning-apples-app-review-process-into-a-nightmare-for-developers.html Comments URL: <a href="https://news.ycombin...

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https://synthanatomy.com/2026/11/ai-is-turning-apples-app-review-process-into-a-nightmare-for-developers.html

Foundation Models framework

👤 kermatt 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://developer.apple.com/foundation-models-framework/ Comments URL: https://news.ycombinator.com/item?id=50474361 Points: 88 # Co...

🔗 **Read full article:**
https://developer.apple.com/foundation-models-framework/

Chinese frontier AI models carry the fingerprint of export-controlled GPUs

👤 firloop 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://unemploymentlabs.com/checkpoint-forensics Comments URL: https://news.ycombinator.com/item?id=50474883 Points: 93 # Comments: 22

🔗 **Read full article:**
https://unemploymentlabs.com/checkpoint-forensics

Why You Should Expect an AI Crash

👤 toomuchtodo 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.economist.com/leaders/2026/11/12/why-you-should-expect-an-ai-crash Comments URL: <a href="https://news.ycombinator.com/item?id=50475735">https://news.ycombinator.com/item?i...

🔗 **Read full article:**
https://www.economist.com/leaders/2026/11/12/why-you-should-expect-an-ai-crash

New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

> **Summary:** We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

Summary: We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

Summary: In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11493624

A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11454471

Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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http://ieeexplore.ieee.org/document/11716332

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📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Front Cover

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11716326

Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361141

Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11359738

Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11364153

Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11359456

Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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http://ieeexplore.ieee.org/document/11359487

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IEEE Transactions on Biomedical Engineering Handling Editors Information

📅 2026-10-21 📖 1 min 📝 1 words TRANSACTIONS BIOMEDICAL ENGINEERING

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IEEE Transactions on Biomedical Engineering Information for Authors

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IEEE Engineering in Medicine and Biology Society Publication Information

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📅 2026-10-21 📖 1 min 📝 1 words TRANSACTIONS BIOMEDICAL ENGINEERING

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11743514

Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

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IEEE Engineering in Medicine and Biology Society

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

Summary: The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

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Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

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EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

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Repetition dissociates pointer and content-based representations in visual working memory: contrasting the CDA with multivariate shape decoding

📖 1 min 📝 16 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Dock H. Duncan, Güven Kandemir, Christian N.L. Olivers

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https://www.sciencedirect.com/science/article/pii/S1053811926006506?dgcid=rss_sd_all

Electroencephalography neurophysiology alterations in association with Aβ and tau pathology across the symptomatic stages of Alzheimer's disease continuum: a cross-sectional study

📖 1 min 📝 32 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Shuo Yang, Zian Pei, Xiaoyong Lan, Chongyuan Lian, Yumei Liu, Zhuofeng Li, Ge Dang, Xue Shi, Huicong Kang, Xin Jiang, Robert Chen, Yi Guo

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The extended default mode network dysconnectivity in schizophrenia

👤 Zarghami, T. S. 📅 2026-11-12 📖 1 min 📝 197 words BIORXIV NEUROSCIENCE

Summary: The default mode network (DMN) extends beyond cortex into subcortical structures, yet how this extended DMN (eDMN) is causally reorganized in schizophrenia (SZ) remains unclear. Here, resting-state fMRI from 74 healthy controls (HC) and 66 SZ patients was analyzed. Subject-specific eDMNs were identi...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765658v1?rss=1

Sleep promotes neuronal autophagy for intracellular clearance

Yang, L., Zhang, X., Li, J., Liu, X., Liu, D.

2026-11-12 1 min 140 words

BIORXIV NEUROSCIENCE

Summary: Sleep is essential for brain health, yet how it maintains neuronal homeostasis at the cellular level remains poorly understood. Here, we show that extended wakefulness selectively suppressed neuronal autophagic flux in the cortex of mice, whereas sleep rapidly restored it, accompanied by a shift in ...

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An AAV model of tauopathy targeting the mouse dentate gyrus reveals greater trans-synaptic tau spreading with aggregation-prone tau variants

Leal, P., Duwat, C., Adamski, C., Gaillard, M.-C., Auregan, G., Guillermier, M., Bernier, S., Dufour, N., Petit, F., Jan, C., Herard, A.-S., Fourmy, D., Kabani, M., Bonvento, G., Cambon, K., Bemelmans, A.-P.

2026-11-12 1 min 297 words

BIORXIV NEUROSCIENCE

Summary: Alzheimer's disease is characterized by progressive memory and cognitive decline, which is strongly correlated with the anatomical progression of tau pathology. Resulting from the intraneuronal aggregation of misfolded tau protein, tauopathy propagates in an anterograde manner between synaptically c...

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MFSD2A, the major transporter of omega-3 DHA in the central nervous system, is expressed in the neural cells of the retina

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77630-2

Social instability distinctly shapes fear memory retention and amygdala function in adolescents and adults

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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Resting plasma irisin shows lower levels in older adults independently of muscle mass and physical activity and is not associated with cognitive performance

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CTNNB1 activation attenuates neuronal apoptosis and improves functional recovery after spinal cord injury via the Wnt/β-catenin signaling pathway

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https://www.nature.com/articles/s41598-026-80398-0

Social determinants of health and the exposome shape aging and dementia across global populations

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s43587-026-01246-4

Correction: Transdiagnostic prevention in youth mental health, Part I: rationale, shared risk factors

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41386-026-02579-9

Dynamic interactions between action generation and body-related multisensory integration

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-79935-8

A prefrontal microcircuit computes trial-wise discount of reinforcement during threat learning in male mice

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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Gene discovery and pleiotropic architecture of multidimensional pain in a genome-wide association study of >1.2 million individuals

Henry R. Kranzler 2026-11-12 1 min 46 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 12 November 2026; doi:10.1038/s41593-026-02455-1 We conducted a meta-analytic genome-wide association study of multidimensional pain in >1.2 million European-ancestry individuals. We identif...

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https://www.nature.com/articles/s41593-026-02455-1

Increased functional connectivity in glioma: separating compensation from tumor-promoting remodeling

Ning Su 2026-11-09 1 min 134 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Increased functional connectivity (FC) in glioma is often interpreted as compensatory neuroplasticity or tumor-promoting circuit remodeling, although FC is a measure of coupling rather than a mechanism. We propose a qualitative framework that first examines whether the measurement is suitable for th...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2005308

Magnetic susceptibility source separation technology: a narrative review of principles, validation, and clinical translation

Lingfei Guo
2026-11-12 1 min 215 words

FRONTIERS NEUROSCIENCE

Summary: Quantitative susceptibility mapping (QSM) returns the net susceptibility of each voxel. When paramagnetic and diamagnetic materials coexist, their opposing contributions can cancel, and QSM cannot recover them separately. Magnetic susceptibility source separation (MSSS) adds a relaxation-sensitive m...

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Accurate EEG-based linear state-space model of cortical dynamics during a passive motor task with exogenous input

Sanna Bakels, Mark van de Ruit and Matin Jafarian
2026-11-12 1 min 205 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Revealing the dynamics of the brain in the presence of exogenous inputs is crucial for mechanistic understanding, interventions and regulations. We study state-space mathematical modeling of brain’s cortical dynamics using electroencephalography (EEG) data in an experiment with continuous...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb97c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Incidence and Management of Ocular Hypertension Following the 0.19 mg Fluocinolone Acetonide Implant for Diabetic Macular Edema

👤 Wills Eye Retina Service Study Group, The Wills Eye Retina Service Study Group additionally includes

📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSION: FAi was associated with low rates of OHT, with most cases managed with observation or topical therapy, with surgery rarely required. This supports the safety of FAi and highlight the importance of extended IOP monitoring.

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Passive imaging with quantum advantage

👤 A I Lvovsky

📅 2026-11-11 📖 1 min 📝 67 words LOW VISION

Summary: Far-field optical imaging inevitably involves low-pass spatial filtering, limiting the resolution. Moreover, conventional imaging suppresses high spatial frequency components close to the cutoff, making them invisible under noise-particularly the shot noise arising from discrete and random nature of...

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Memory-guided mutual information optimization in retinal imaging for early stroke detection and subtyping

👤 Yitian Zhao 📅 2026-11-11 📖 1 min 📝 64 words LOW VISION

Summary: Effective stroke prevention relies on identifying at-risk individuals before neurological symptoms arise, yet this is severely limited by the absence of scalable screening tools. To address this, we developed RetCEVD, a diagnostic framework that uses rapid, non-invasive retinal Optical Coherence Tom...

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SnowRef-Drive: A Global Instruction-Driven Traffic Video Segmentation Dataset for Adverse Weather Driving Scenarios

👤 Yintao Wei 📅 2026-11-11 📖 1 min 📝 62 words LOW VISION

Summary: Adverse weather, particularly snowfall and low visibility, severely degrades vision-based traffic perception, yet existing language-driven video segmentation datasets rarely provide pixel-level annotations for real-world driving under such conditions. We present SnowRef-Drive, a large-scale global d...

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Predictors of visual acuity change in a large cohort study of scleritis from Northeast China

👤 Xiaoli Liu
📅 2026-11-11 📖 1 min 📝 39 words

LOW VISION

Summary: CONCLUSION: This Northeast Chinese scleritis cohort is predominantly anterior and female-predominant, with low rates of bilateral involvement, necrotizing type, and associated autoimmune diseases. Furthermore, this study indicates the predictors of visual acuity change, providing reference points fo...

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Comparison of outcome in patients with idiopathic intracranial hypertension (IIH) following optic nerve sheath fenestration (ONSF) using a medial transconjunctival approach with dural window or superomedial lid-crease incision approach

👤 Philip A Tenzel
📅 2026-11-12 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: This study demonstrates that the superomedial lid-crease approach with fenestration and the medial transconjunctival approach with dural window for ONSF are both effective in preventing vision loss in patients with IIH, with similarly low side-effect profiles.

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Geographic concordance of cataract and glaucoma burden and associated country-level exposures across 204 countries and territories

Changzheng Chen 2026-11-12 1 min 60 words LOW VISION

Summary: CONCLUSION: National cataract- and glaucoma-related vision-loss burdens showed geographic concordance. The exposure findings were country-level associations, and their statistical support varied across regional and spatial models. These geographic patterns may help inform planning for cataract and g...

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Dynamic receptive field and feature enhanced network for bridge defect detection

Hui Li 2026-11-12 1 min 50 words LOW VISION

Summary: INTRODUCTION: Accurate defect detection in underwater and cross-domain marine infrastructure, such as bridge piers, submarine pipelines, and subsea equipment, is essential for structural safety assessment and intelligent ocean engineering. Existing methods, however, suffer from large scale variation...

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Demographic and geographic variations in US blindness and low vision mortality, 1999-2020

👤 Aalaa Saleh 📅 2026-11-12 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSION: Blindness- and low-vision-related mortality in the United States exhibit strong age dependence and persistent demographic and geographic disparities. The recent stagnation in the decline in mortality highlights the need for targeted, equity-focused public health interventions to reduce p...

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Influencing Factors of Positional Stability of Implantable Collamer Lenses in Terms of Rotation, Decentration, and Tilt in Patients With Myopia

👤 Jing Zhao 📅 2026-11-12 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSIONS: ICL positional stability was within an acceptable range. ICL fixation angle and the structures of the iris and ciliary body influenced the rotation, decentration, and tilt of the ICL. Abnormal vault values were also correlated with ICL rotation and tilt.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

Paul M Briley 2026-11-10 1 min 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

👤 Hamed Ekhtiari 📅 2026-11-10 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

👤 Shouping Zhao 📅 2026-11-11 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

👤 Lei Huang 📅 2026-11-11 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

👤 Zahra Hadi Alqallaf 📅 2026-11-11 📖 1 min 📝 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

👤 Kanecia O Zimmerman 📅 2026-11-11 📖 1 min 📝 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Post-stroke functional independence after brain stimulation: meta-analysis

Sevda Uzun 2026-11-12 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSION: The findings suggest that transcranial direct current stimulation may contribute to improvements in motor functional independence following stroke. However, further studies with standardized intervention protocols and larger sample sizes are needed to confirm these findings.

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou
2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

👤 Zhanhua Liang
📅 2026-11-11 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

👤 Quanyan Zhu 📅 2026-11-10 📖 1 min 📝 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

👤 Shuang Wang 📅 2026-11-10 📖 1 min 📝 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger
2026-11-11 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this animal, condition-dependent dimensionality was significantly higher in F5 than in AIP or M1, while alignment and reconstruction analyses showed descriptive area-specific patterns in how closely combined-finger activity resembled a simple combination of single-finger activity. For...

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Effects of brain-computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

👤 Lili Fang 📅 2026-11-11 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Brain-computer interface-triggered functional electrical stimulation showed a clearer signal of benefit vs. conventional rehabilitation than brain-computer interface-controlled robotic. However, the comparison between the two active modalities relies entirely on indirect evidence from a ...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

👤 Jilai Li 📅 2026-11-11 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Spinal cord-computer interfaces for neurorehabilitation: Technical challenges and clinical perspectives

Minfei Wu 2026-11-11 1 min 35 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: This Perspective aims to clarify the clinical and engineering scope of spinal cord-computer interfaces (SCCIs) in neurorehabilitation, distinguishing them from adjacent brain-computer interface, neuromodulation, and assistive-technology approaches while identifying the key barriers to tra...

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Clinical safety assessment of implanted brain-computer interfaces: current evidence and a pathway to standardization

Wen-Jun Tu 2026-11-12 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Implanted brain-computer interfaces are beginning to move beyond short laboratory sessions into long-term and home use. The evidence used to judge clinical safety remains fragmented. This review examines penetrating intracortical, cortical-surface (including subpial) and cerebral-venous systems, wit...

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

📅 2024-01-22 📖 1 min 📝 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition's foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 2 min 📝 586 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

112 CodeQL alerts on my python app, 14 were real

👤 /u/Stunning-Sherbet1853 📅 2026-11-12 📖 1 min 📝 398 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> HI, i run a long-lived python backend on windows and mac! it only listens on 127.0.0.1 and almost every route needs an access token. last week i turned on the default CodeQL scan for the first time and got 112 alerts (112 → 1 ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xujdbk/112_codeql_alerts_on_my_python_app_14_were_real/

ExcalidrawZ: Local Native Excalidraw on macOS/iOS

👤 Qision 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/chocoford/ExcalidrawZ Comments URL: https://news.ycombinator.com/item?id=50468944 Points: 61 # Comments: 18

🔗 **Read full article:**
https://github.com/chocoford/ExcalidrawZ

Nintendo Takes Down Emulators and Firmware in Massive GitHub DMCA Sweep

👤 CharlesW 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.ign.com/articles/nintendo-takes-down-switch-emulators-and-mod-chip-firmware-in-massive-github-dmca-sweep Comments URL: <a href="https:...

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https://www.ign.com/articles/nintendo-takes-down-switch-emulators-and-mod-chip-firmware-in-massive-github-dmca-sweep

AWS Lambda Web Functions keep billing memory ~7 minutes after the last request

👤 seanw265 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.easel.sh/blog/lambda-web-functions-idle-memory Comments URL: https://news.ycombinator.com/item?id=50470445 Points: 38 ...

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https://www.easel.sh/blog/lambda-web-functions-idle-memory

Paths in Pieces

👤 acossta 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://stoppels.ch/2026/11/12/paths-in-pieces.html Comments URL: https://news.ycombinator.com/item?id=50470762 Points: 13 # Comments: 0<...

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https://stoppels.ch/2026/11/12/paths-in-pieces.html

Show HN: Kapso – Iterative learning from enterprise data

👤 alireza_msh 📅 2026-11-12 📖 4 min 📝 826 words HACKER NEWS

> **Summary:** Hey HN! We're Alireza and Majid building Leeroo (https://leeroo.com/). Previously we've built AI systems for products with millions of users, including at LinkedIn and Meta. We're now focusing on building AI systems for enterprise data. Today, we're open-sourcing...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50471281

Gemini 4 Hydrogen

👤 dvaplima 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.google/innovation-and-ai/models-and-research/gemini-models/introducing-gemini-4-hydrogen/ Comments URL: <a href="https://news.ycombinator.com/item?...

🔗 **Read full article:**
https://blog.google/innovation-and-ai/models-and-research/gemini-models/introducing-gemini-4-hydrogen/

Show HN: AgentRun – open-source, Self-improving Managed Agents

👤 miguelrios 📅 2026-11-12 📖 1 min 📝 133 words HACKER NEWS

> **Summary:** We (GREP.AI, YC F26) are open-sourcing AgentRun and turning it into a full managed agents platform. Read more about it in this post: <a href="https://agentrun.ai/blog/introducing-agentrun-run-smarter-agents-use-fewer-tokens" rel="nofollow">https://agentrun.ai/blog/introducing-agentrun-run-smarter...

🔗 **Read full article:**
https://github.com/Parcha-ai/agentrun/blob/main/README.md

Ask HN: Microsoft/Azure Outage?

👤 nazgulsenpai 📅 2026-11-12 📖 1 min 📝 32 words HACKER NEWS

> **Summary:** For the past 5 minutes all of our Microsoft Cloud connections are down. Anyone else? The Status page won't even load. <hr /> Comments URL: https://news.ycombinator.com/item?id=50471668 Points: 64 # Comments: 22

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50471668

Outlook is down, and so is Microsoft's status page

👤 dkpk 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gvwire.com/2026/11/12/microsoft-outlook-down-for-thousands-of-users-downdetector-shows/ Comments URL: <a href="https://news.ycombinator.com/item?id=50471731">h...

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Rust 1.100.0

👤 thesuperbigfrog 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/rust-lang/rust/releases/tag/1.100.0 Comments URL: https://news.ycombinator.com/item?id=50471793 Points: 50 # Commen...

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https://github.com/rust-lang/rust/releases/tag/1.100.0

RAD51 expression in pan-cancer and its prognostic and immune mechanisms in glioma

📖 1 min 📝 35 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Liubing Hou, Yu Wang, Xiang Zhan, Jiayuan Li, Zizhou Zhang, Weijing Cai, Linlin Su, Wei Han, Xuetao Han, Yue Han, Xiaolan Lv, Xiaoying Xue, Huandi Zhou

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004816?dgcid=rss_sd_all

Reduced hypothalamic [³H]PK11195 binding in Lewis rats after chronic high-intensity interval training

📖 1 min 📝 28 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Atiqur Hasany, John R. Tyler, Huy Lu, Kenneth Blum, Wei Sun, Richard J. Salvi, Igor Elman, Panayotis K. Thanos

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004622?dgcid=rss_sd_all

Alpha-beta oscillatory dynamics underlying auditory inhibitory control in children

Finnilä, E., van Bijnen, S., Kujala, J., Parviainen, T.

2026-11-11 1 min 180 words

BIORXIV NEUROSCIENCE

Summary: Response inhibition, the ability to withhold an unwanted response, is a fundamental component of cognitive control. While evoked responses in inhibitory tasks have been widely studied, induced oscillatory dynamics, particularly in the auditory modality and in children, remain largely unexplored. Her...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765597v1?rss=1

How striatal neurons influence decision-making depends on broader circuit context: evidence from theory-driven cell type identification

Yu, Z., Verstynen, T., Rubin, J.

2026-11-11 1 min 249 words

BIORXIV NEUROSCIENCE

Summary: Neural cell types are often conceived as having fixed computational roles, yet their behavioral contributions may depend on the broader circuit state. In the striatum, for example, direct and indirect pathway spiny projection neurons (SPNs) are typically described as facilitating or suppressing acti...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765679v1?rss=1

Postpartum resource scarcity reshapes maternal care, offspring prioritization, and the steroid-sensitive medial preoptic area

👤 Aspesi, D., Soriano, V., Rolle, S., Mahrous, M. Y., Brenhouse, H., Krishnan, K., Bangasser, D. A.

📅 2026-11-12 📖 1 min 📝 249 words BIORXIV NEUROSCIENCE

Summary: Parental care requires balancing offspring needs with environmental constraints, yet how resource scarcity alters the organization of maternal responses remains poorly understood. Here, we examined maternal behavior, offspring prioritization, and steroid receptor-associated extracellular matrix orga...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765420v1?rss=1

Sentence-level neural dynamics in speech motor cortex

Levi-Aharoni, H., Jude, J. J., Haro, S., Acosta, A. J., Allcroft, S. B., Ottenhoff, M. C., Card, N. S., Wairagkar, M., Iacobacci, C., Pritchard, A. L., Richards, B., Brandman, D. M., Stavisky, S. D., Au Yong, N., Pandarinath, C., Paulk, A. C., Hadar, P. N., Li, J., Williams, Z. M., Cash, S. S., Simeral, J. D., Hochberg, L. R., Rubin, D. B.

2026-11-12 1 min 155 words BIORXIV NEUROSCIENCE

Summary: Prior recordings from speech motor cortex have revealed phonemic, syllabic, articulatory and speech-sequence representations, while more recent studies indicate that activity may also reflect higher-order features of speech production. However, whether speech motor cortex is sensitive to the linguis...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765804v1?rss=1

The Computational Grammar of Human Behaviour in a Biological Environment

Serratos Hernandez, U. D., Goekay, U., Kornemann, L., Hutabarat, Y., Sen, D., Bach, D. R.

2026-11-12 1 min 193 words BIORXIV NEUROSCIENCE

Summary: Human behaviour unfolds as an extended stream of spontaneous actions shaped by environmental demands and the agent's goals. The organisation of these actions, situated between setting abstract goals and concrete movement execution, remains poorly characterised. Here, we introduce DynaClade, an unsup...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765265v1?rss=1

Loss of GluN2A Selectively Impairs Excitatory Synaptic Integration in Hippocampal Somatostatin Interneurons

Hunter, D. W., Kaese, H., Kind, P. C., Wyllie, D. J.
2026-11-12 1 min 224 words

BIORXIV NEUROSCIENCE

Summary: N-methyl-D-aspartate receptors (NMDARs) are widely considered key determinants of excitatory synaptic kinetics and activity-dependent plasticity, yet their specific contribution to postsynaptic potentials across distinct hippocampal CA1 cell types remains poorly defined. Null variants in GRIN2A are ...

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Same Movements, Different Skills: How Context Shapes Learning in Humans and Neural Networks

Gupta, M. W., Taylor, J. A.
2026-11-12 1 min 149 words

BIORXIV NEUROSCIENCE

Summary: What determines whether experiences are integrated into shared representations or split? Motor skill learning offers a revealing test: when two tasks require an identical sequence of movements, both intuition and theory suggest that practice in one should generalize to the other. Yet we find that ch...

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A hypothesis-generating multi-resolution post-GWAS analysis of context-dependent regulatory signals in migraine

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-79784-5

Alterations in glymphatic system in patients with Sleep-related epilepsy: a study of PVS and DTI-ALPS index

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Bridging the dementia diagnostic gap by addressing cognitive assessments globally

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NeuroMemory-DiffNet: a reference-guided memory framework for brain tumor segmentation and classification

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Beat perception relates to reading through phonological awareness and rapid automatized naming

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https://www.nature.com/articles/s41598-026-72872-6

When neuroscience meets machine learning: a taxonomic scoping review of machine and statistical learning approaches in sub-Saharan Africa

Deshendran Moodley
2026-11-12 1 min 146 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: In high-resource economies, computational neuroscience research frequently employs machine learning (ML) and statistical learning (SL) techniques. While these methods provide effective ways to analyse multimodal, high-dimensional, and complex data they are underexplored in low resource settings. Thi...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1975649

Validation of the Hungarian version of the Coma Recovery Scale-Revised in patients with disorders of consciousness

András Attila Horváth
2026-11-12 1 min 171 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveOur objective was to validate the Hungarian version of the Coma Recovery Scale-Revised (CRS-R).DesignWe conducted a multicenter study across neurology, intensive care, neuro-rehabilitation, and nursing departments in Hungary. Thirty-one patients (17 males; mean age 49.1 ± 16.1 years) with d...

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Coordination without command: active inference and the route to embodied intelligence

Alexander D. Shaw 2026-11-12 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: Embodied intelligence depends on bodies whose contact with the environment, distribution of load and capacity for action are only partly known and are changed by movement. We argue that this creates a practical reason to organize inference through distributed, hierarchical and plural processes, usin...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1898490

Case Report: Clinical applications of extended-duration remote ambulatory EEG monitoring: lessons from real-world cases

Timothy Ryan 2026-11-12 1 min 191 words FRONTIERS NEUROSCIENCE

Summary: BackgroundTraditional ambulatory electroencephalography (EEG) and inpatient epilepsy monitoring unit (EMU) admissions face significant operational limitations including: short recording windows (typically 24–72 h), high costs, technician shortages, and physical restrictions on daily activities.Metho...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1976990

What shared responses reveal about audience behavior: neural prediction and the emerging potential of peripheral synchrony

Isak B. Blank 2026-11-12 1 min 214 words FRONTIERS NEUROSCIENCE

Summary: Predicting audience responses to dynamic media remains a significant challenge in neuromarketing and neuroforecasting. Intersubject correlation (ISC) has been proposed to address this need by quantifying the temporal similarity of neural, physiological, and behavioral responses among individuals exp...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ageist attitudes as a moderator of objective and subjective vision: A longitudinal study

Yoav S Bergman 2026-11-11 1 min 41 words LOW VISION

Summary: OBJECTIVES: This longitudinal study examined the relationship between objective visual acuity and subjective vision over a six-month period. Specifically, it investigated the moderating role of internalized ageist attitudes in shaping how objective vision relates to perceived sensory vision difficul...

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Registry-recorded stroke and transient ischaemic attack diagnoses in Saudi Arabia, 2016-2025: a national descriptive analysis

👤 Mohammed AlAbdulali
📅 2026-11-11 📖 1 min 📝 58 words

LOW VISION

Summary: CONCLUSION: Saudi Arabia's recorded stroke burden is concentrated among older males. The findings describe patterns captured by the PHO and should not be interpreted as prevalence, incidence, or causal associations, which indicate the need for better prevention, management of chronic diseases and a ...

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Design and parameter optimization of a bilateral staggered wheel-spoon precision seed-metering device for millet plot breeding

👤 Ankang Mu 📅 2026-11-11 📖 1 min 📝 63 words LOW VISION

Summary: Precision seeding of micro-sized millet seeds in plot breeding faces critical challenges, including high miss-seeding rates and severe seed cross-contamination during variety changeovers. This study developed a novel bilateral staggered wheel-spoon precision seed-metering device to achieve precise h...

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Representation geometry and task performance in vision-language modeling for computed tomography enterography

👤 Ryan Stidham 📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSIONS: Slice sampling and tissue-contrast-focused encoding affect performance in CT enterography vision-language systems, and RAG improved report-label agreement over the fine-tuned baseline. These results provide initial baselines and practical guidance for multimodal learning in volumetric C...

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Incidence and Management of Ocular Hypertension Following the 0.19 mg Fluocinolone Acetonide Implant for Diabetic Macular Edema

👤 Wills Eye Retina Service Study Group, The Wills Eye Retina Service Study Group additionally includes

📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSION: FAi was associated with low rates of OHT, with most cases managed with observation or topical therapy, with surgery rarely required. This supports the safety of FAi and highlight the importance of extended IOP monitoring.

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Passive imaging with quantum advantage

👤 A I Lvovsky

📅 2026-11-11 📖 1 min 📝 67 words LOW VISION

Summary: Far-field optical imaging inevitably involves low-pass spatial filtering, limiting the resolution. Moreover, conventional imaging suppresses high spatial frequency components close to the cutoff, making them invisible under noise-particularly the shot noise arising from discrete and random nature of...

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Memory-guided mutual information optimization in retinal imaging for early stroke detection and subtyping

👤 Yitian Zhao 📅 2026-11-11 📖 1 min 📝 64 words LOW VISION

Summary: Effective stroke prevention relies on identifying at-risk individuals before neurological symptoms arise, yet this is severely limited by the absence of scalable screening tools. To address this, we developed RetCEVD, a diagnostic framework that uses rapid, non-invasive retinal Optical Coherence Tom...

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SnowRef-Drive: A Global Instruction-Driven Traffic Video Segmentation Dataset for Adverse Weather Driving Scenarios

👤 Yintao Wei 📅 2026-11-11 📖 1 min 📝 62 words LOW VISION

Summary: Adverse weather, particularly snowfall and low visibility, severely degrades vision-based traffic perception, yet existing language-driven video segmentation datasets rarely provide pixel-level annotations for real-world driving under such conditions. We present SnowRef-Drive, a large-scale global d...

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Predictors of visual acuity change in a large cohort study of scleritis from Northeast China

Xiaoli Liu
2026-11-11 1 min 39 words LOW VISION

Summary: CONCLUSION: This Northeast Chinese scleritis cohort is predominantly anterior and female-predominant, with low rates of bilateral involvement, necrotizing type, and associated autoimmune diseases. Furthermore, this study indicates the predictors of visual acuity change, providing reference points fo...

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Comparison of outcome in patients with idiopathic intracranial hypertension (IIH) following optic nerve sheath fenestration (ONSF) using a medial transconjunctival approach with dural window or superomedial lid-crease incision approach

👤 Philip A Tenzel
📅 2026-11-12 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: This study demonstrates that the superomedial lid-crease approach with fenestration and the medial transconjunctival approach with dural window for ONSF are both effective in preventing vision loss in patients with IIH, with similarly low side-effect profiles.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

👤 Paul M Briley 📅 2026-11-10 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

👤 Hamed Ekhtiari 📅 2026-11-10 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

Shouping Zhao 2026-11-11 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

Lei Huang 2026-11-11 1 min 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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MSTA-CNN an ultra-lightweight multi-scale temporal-spatial attention convolutional neural network for high-performance P300 spellers

👤 Mousa Shamsi
📅 2026-11-10 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Efficient P300 detection in brain-computer interfaces (BCIs) remains a challenge due to low signal-to-noise ratio (SNR) and the high computational cost of existing deep learning models, which limits their deployment in real-time applications. To address these issues, we propose MSTA-CNN, an ultra-li...

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger
2026-11-11 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this animal, condition-dependent dimensionality was significantly higher in F5 than in AIP or M1, while alignment and reconstruction analyses showed descriptive area-specific patterns in how closely combined-finger activity resembled a simple combination of single-finger activity. For...

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Effects of brain-computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

Lili Fang 2026-11-11 1 min 54 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Brain-computer interface-triggered functional electrical stimulation showed a clearer signal of benefit vs. conventional rehabilitation than brain-computer interface-controlled robotic. However, the comparison between the two active modalities relies entirely on indirect evidence from a ...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

Jilai Li 2026-11-11 1 min 2 words BRAIN COMPUTER INTERFACE

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Spinal cord-computer interfaces for neurorehabilitation: Technical challenges and clinical perspectives

Minfei Wu 2026-11-11 1 min 35 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: This Perspective aims to clarify the clinical and engineering scope of spinal cord-computer interfaces (SCCIs) in neurorehabilitation, distinguishing them from adjacent brain-computer interface, neuromodulation, and assistive-technology approaches while identifying the key barriers to tra...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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http://doi.org/10.1037/neu0001096

Critical RCE Vulnerability in Apache HTTP Server

👤 soheilpro 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

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Lean Grimoire

👤 smilelamp 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

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Darling Development Update – Q4 2026

👤 tech234a 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.darlinghq.org/2026/11/08/development-update-q4-2026/ Comments URL: https://news.ycombinator.com/item?id=50463103 Poin...

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I'm Working on a FOSS Linux Pinbal Simulation Engine

👤 sysrpl 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

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Clip-to-trial: EEG-based decoding network with long short-term self-attention for DRDS recognition

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Lili Shen, Yiming Wu, Zihao Yang, Xinyuan Di

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007074?dgcid=rss_sd_all

Intranasal insulin improves long-term cognitive deficits and protects hippocampal damage following neonatal hypoxia–ischemia in juvenile rats

📖 1 min 📝 41 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Nilesh Dankhara, Jonathan W. Lee, Norma B. Ojeda, Michelle A. Tucci, Silu Lu, Elizabeth L. White, Lindsey E. Winborne, Sharon Xu, Carolyn K. Glendye, Lu-Tai Tien, Yi Pang, Abhay J. Bhatt, Lir-Wan Fan</...

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https://www.sciencedirect.com/science/article/pii/S0306452226007153?dgcid=rss_sd_all

Transcriptional and DNA Methylation Changes in the Hippocampus and Prefrontal Cortex after Fractionated Cranial Radiation in Mice

👤 Aksoy, T., Sumazin, P., Lu, Y., Najaf Panah, M. J., Zamorano, M., Lam, T., Zhou, W., Veselinovic, I., Duman, J. G., Gaber, M. W., Heijnen, C. J., Grosshans, D. R.

📅 2026-11-11 📖 1 min 📝 163 words BIORXIV NEUROSCIENCE

Summary: Fractionated whole-brain irradiation is essential for treating brain tumors, but many long-term survivors develop cognitive decline whose molecular basis is poorly understood. We asked which cell types and gene programs change late after irradiation, how the hippocampus and prefrontal cortex compare...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765716v1?rss=1

A common neural axis links image distinctiveness and memorability

👤 Utochkin, I., Chang, W., Vogel, E. K., Awh, E.

📅 2026-11-11 📖 1 min 📝 139 words

BIORXIV NEUROSCIENCE

Summary: Stimulus-specific memorability, the tendency for people to remember and forget the same stimuli, is a powerful predictor of memory performance that is independent of the study and retrieval factors that have dominated memory research. Critically, memorability can be defined in two ways: "Hit-ability...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765672v1?rss=1

Neuropathological assessment of cerebrovascular integrity in Chronic Traumatic Encephalopathy

👤 Martinac, A. E., DeLorenzo, A., Dik, D., Pasas, L., Bilston, L. E., Shepherd, C. E.

📅 2026-11-11

📖 1 min 📝 245 words

BIORXIV NEUROSCIENCE

Summary: Chronic traumatic encephalopathy neuropathological change (CTE-NC) is defined by perivascular deposits of phosphorylated tau, with lesions preferentially located at sulcal depths. We investigated whether CTE-NC is associated with structural changes to perivascular spaces (PVSs) or microvascular dege...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765492v1?rss=1

Distinct input-output relationships between basal forebrain subnuclei underlie heterogeneous cholinergic modulation of the mouse cortex

👤 Pinto, L., Tan, J., Harter, A. M., Kolb, M., Lu, R.

📅 2026-11-11 📖 1 min 📝 150 words

BIORXIV NEUROSCIENCE

Summary: The basal forebrain (BF) supports many behavioral functions via its brain-wide projections, which can be recruited independently and task specifically. However, the anatomical underpinnings for this functional heterogeneity are poorly understood. Here, we compared two mouse BF subnuclei, the horizon...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765460v1?rss=1

Striatal dopamine availability peaks in adulthood and undergoes age-dependent modulation

White, S. R., Shin, J. H., Goldbach, H., Andujar, M., Swanson, E. S., Rimondini, R., Carr, A. L., Anderson, L. G., Cerveny, S. E., Harmon, P. Q., Tangen, A. C., Pelkey, K. A., Azadi, R., Cummins, A. C., McBain, C. J., Averbeck, B., Alvarez, V. A.

2026-11-11 1 min 71 words BIORXIV NEUROSCIENCE

Summary: Age is a risk factor for several psychiatric disorders. An adolescent peak in striatal dopamine availability is thought to contribute to the neuropsychiatric vulnerability. However, the lifespan trajectory of striatal dopamine release capacity is undefined in most species. Here, we measured lifespan...

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Advanced meditation stabilizes cortical dynamics through inhibitory gain control linked to absorptive depth

Panda, R., Kumar, G., Nunez, P., Subramaniam, B., Bhandare, K., Malipeddi, S., Major, G., Gosseries, O., Laureys, S., Tewarie, P.

2026-11-11 1 min 256 words BIORXIV NEUROSCIENCE

Summary: Advanced meditation can shift conscious experience from ordinary body- and world-centered awareness toward absorption, detachment and dissociation-like states, yet the neural mechanisms that stabilize these immersive states remain unclear. We studied 18 long-term meditators during eyes-open rest and...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.08.765124v1?rss=1

Learned associations constrain anticipatory visual dynamics and enhance sensory and motor readiness to facilitate perceptual decisions

Zhang, L., Yuan, M., Zhang, H., Morris Moscovitch, M., Alain, C.

2026-11-11 1 min

246 words

BIORXIV NEUROSCIENCE

Summary: Learned cue target associations can guide attention to facilitate perceptual decision making, yet how they shape the content and dynamics of neural activity, and whether this facilitation reflects anticipatory preparation, post-target sensory enhancement, or both, remains unclear. Here, we used magn...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765786v1?rss=1

Tri-Modal Single-Cell Atlas of Developing Human Brain

Song, L., Wang, Z., Zhu, B., DeKuyper, F. A., Wang, C., Liao, Z., Girdhar, K., Lee, D., Tsankova, N. M., Kaji, D. A., Fullard, J. F., Dong, P., Roussos, P., Ma, S.

2026-11-11 1 min

139 words

BIORXIV NEUROSCIENCE

Summary: Epigenetic modifications, such as DNA methylation, evolve throughout development and aging, providing remarkable regulatory diversity and contributing to disease etiology, including neuropsychiatric disorders. However, how these regulatory layers interact across human brain development remains uncle...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.10.765750v1?rss=1

Hyperglycemia-responsive circRNA drives diabetic neuropathic pain by competing for PUM2 binding and promoting proNT3/CCL2 in DRG neurons

👤 Li, X., Huang, C., Govindarajalu, G., Wang, B., Zhang, H.-H., Feng, X., Sharma, D., Ma, R., Zhang, L., Zheng, B., Xiao, J., Ferreira, M. d. A., Berkman, T., Cabrera, J., Li, X., Jiang, B.-C., Zhang, X.-S., Davidson, S., TAO, Y.-X.

📅 2026-11-11 📖 1 min 📝 199 words BIORXIV NEUROSCIENCE

Summary: Diabetic neuropathic pain (DNP), one of the most common and debilitating complications of diabetes mellitus (DM), remains poorly managed by conventional analgesics, highlighting the urgent need to explore its etiology-associated mechanisms. Here, we identified a novel conserved circular RNA, cRps6kb...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765753v1?rss=1

A novel behavioral paradigm for characterizing anticipatory postural adjustments in mice

👤 Morina, E., Claudi, F., Murray, A. J.
📅 2026-11-11 📖 1 min 📝 274 words

BIORXIV NEUROSCIENCE

Summary: Voluntary movements perturb the body's equilibrium, and the nervous system limits this disturbance by generating anticipatory postural adjustments (APAs) that redistribute body weight before and during movement. APAs have been characterized extensively in humans and larger animal models, but the abs...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765800v1?rss=1

RNA interference as a precision therapeutic platform for Parkinson’s disease: from molecular mechanisms to translational challenges

📅 2026-11-12 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44573-026-00028-8

Explainable AI-driven neural activity analysis in Parkinsonian rats under electrical stimulation

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79944-7

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

👤 Yingjie Zhu 📅 2026-11-03 📖 1 min 📝 24 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 03 November 2026; doi:10.1038/s41593-026-02535-2 Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02535-2

Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

Ole A. Andreassen
2026-11-06 1 min 25 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 06 November 2026; doi:10.1038/s41593-026-02536-1 Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02536-1

Spatial transcriptomics reveals organizational properties of the mouse spinal cord and alterations in neuropathic pain

Qimeng WangYunfei HuYuling ZhuGautham ChelliahJing PengEmmanuella Osei-AsanteMelissa SobankoBrianna SanchezDavid D. GintyXin Maizie ZhouShan Meltzerahttps://ror.org/02vm5rt34Department of Pharmacology, Vanderbilt University, Nashville, TN 37240bhttps://ror.org/02vm5rt34Department of Computer Science, Vanderbilt University, Nashville, TN 37235chttps://ror.org/02vm5rt34Department of Biomedical Engineering, Vanderbilt University, Nashville, TN 37235dDepartment of Neurobiology, HHMI, Harvard Medical School, Boston, MA 02115

2026-11-10 1 min 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 46, November 2026.
SignificanceThe spinal cord receives and integrates somatosensory inputs and executes motor and autonomic outputs, yet the spatial organization of its diverse cell types in the spinal cord remains incompletel...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2623177123?af=R

Meditation as a bioactive intervention: evidence of translational potential via Connectivity Mapping and TMS neurophysiology

Mitali Mukerji 2026-11-09 1 min 295 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMeditation practices are often used as non-pharmacological adjunct therapy for managing stress and wellbeing benefits. Here, we investigate whether molecular signatures via meditation practices have potential overlap with those of known drug targets that have established neurophysiological...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1904068

Hearing loss as an inference problem: sensory precision, conversational repair, and partner adaptation

Sebastian Schnettler 2026-11-09 1 min 289 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conversational understanding is often approached as a problem of successful signal transmission. This paper develops a complementary account in which conversation is treated as a reciprocal process of inference and action: interlocutors must infer the latent causes of one another’s signals, estimate...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1963610

Noise resilience in brain decoding driven by spiking neural networks for responsive robot control

👤 Alessandro De Luca
📅 2026-11-09 📖 1 min 📝 291 words

FRONTIERS NEUROROBOTICS

Summary: Brain computer interface (BCI) is a promising technology, expanding on many fields ranging from medical to commercial applications. The diffusion of this technique is compromised by the complexity of the treated data and the unavoidable presence of noise in those systems. This study contributes to t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1931325

Transformer networks are robust learners of source localization for EEG measurements

👤 David Hart, Simon Polishchuk and Moritz Dannhauer
📅 2026-11-06 📖 1 min 📝 356 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade un...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb82c

ACF-Net: design of an adaptive cross-modal fusion network for cross-subject wearable EEG-EMG gesture recognition

TongLiang Yang
2026-11-06 1 min 292 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Cross-subject gesture recognition in wearable brain–computer interfaces is constrained by four structural deficiencies: symmetric cross-modal attention discarding cortical precedence over muscular execution, priors confined to post hoc analysis, invariance anchored on unimodal features, a...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb723

The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Alon Gorenshtein, Yosef Adiniaev, Tom Liba, Yiftach Barash, Eyal Klang and Oved Daniel

2026-11-06 1 min 297 words JOURNAL NEURAL ENGINEERING

Summary: Objective. A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts. Approach. Retrospective secondary analysis of BigP3BCI 1...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb62c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Segmentation, Reconstruction, and Grasp-Pose Estimation Using an RGB-D Camera for Robotic Apple Harvesting

Jianqiang Lu 2026-11-10 1 min 58 words LOW VISION

Summary: Accurate segmentation, three-dimensional reconstruction, and grasp-pose estimation of fruits and surrounding structures are essential for robotic harvesting in complex orchard environments. However, occlusions caused by branches, support wires, and irregular canopy structures remain major challenges...

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Intelligent rendering optimization for animated scenes using reinforcement learning and computer vision

Li Jia 2026-11-10 1 min 63 words LOW VISION

Summary: Real-time rendering of complex animated scenes is computationally intensive and often struggles to balance visual fidelity and performance in dynamic or densely populated environments, as conventional pipelines lack adaptive resource optimization, leading to reduced frame rates and degraded visual q...

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Clinical efficacy of unilateral biportal endoscopic spinal decompression for lumbar intraspinal pneumatosis

Huiqiu Jian 2026-11-10 1 min 48 words LOW VISION

Summary: CONCLUSION: UBE for the treatment of lumbar intraspinal pneumatosis has the advantages of minimal trauma, clear field of vision, flexible operation, and rapid postoperative recovery, representing a potentially safe and effective option. Large-scale multicenter prospective trials are still needed to ...

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Photobleaching-Driven Ratiometric Fluorescent Switching of ZGO:Mn Nanorods/RhB for Iodine Vapor Monitoring with Integrated Gas-Phase Separation

Kui Zhang 2026-11-11 1 min 56 words LOW VISION

Summary: Conventional fluorescent sensors generally rely on complex, time-consuming structural engineering of recognition moieties. By contrast, leveraging photobleaching for eco-friendly, reagent-free signal transduction has long been impeded by irreversible fluorophore degradation. Additionally, direct liq...

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Ageist attitudes as a moderator of objective and subjective vision: A longitudinal study

Yoav S Bergman 2026-11-11 1 min 41 words LOW VISION

Summary: OBJECTIVES: This longitudinal study examined the relationship between objective visual acuity and subjective vision over a six-month period. Specifically, it investigated the moderating role of internalized ageist attitudes in shaping how objective vision relates to perceived sensory vision difficul...

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Registry-recorded stroke and transient ischaemic attack diagnoses in Saudi Arabia, 2016-2025: a national descriptive analysis

Mohammed AlAbdulali 2026-11-11 1 min 58 words LOW VISION

Summary: CONCLUSION: Saudi Arabia's recorded stroke burden is concentrated among older males. The findings describe patterns captured by the PHO and should not be interpreted as prevalence, incidence, or causal associations, which indicate the need for better prevention, management of chronic diseases and a ...

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Design and parameter optimization of a bilateral staggered wheel-spoon precision seed-metering device for millet plot breeding

👤 Ankang Mu 📅 2026-11-11 📖 1 min 📝 63 words LOW VISION

Summary: Precision seeding of micro-sized millet seeds in plot breeding faces critical challenges, including high miss-seeding rates and severe seed cross-contamination during variety changeovers. This study developed a novel bilateral staggered wheel-spoon precision seed-metering device to achieve precise h...

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Representation geometry and task performance in vision-language modeling for computed tomography enterography

👤 Ryan Stidham 📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSIONS: Slice sampling and tissue-contrast-focused encoding affect performance in CT enterography vision-language systems, and RAG improved report-label agreement over the fine-tuned baseline. These results provide initial baselines and practical guidance for multimodal learning in volumetric C...

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Incidence and Management of Ocular Hypertension Following the 0.19 mg Fluocinolone Acetonide Implant for Diabetic Macular Edema

👤 Wills Eye Retina Service Study Group, The Wills Eye Retina Service Study Group additionally includes

📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSION: FAi was associated with low rates of OHT, with most cases managed with observation or topical therapy, with surgery rarely required. This supports the safety of FAi and highlight the importance of extended IOP monitoring.

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Passive imaging with quantum advantage

👤 A I Lvovsky

📅 2026-11-11 📖 1 min 📝 67 words LOW VISION

Summary: Far-field optical imaging inevitably involves low-pass spatial filtering, limiting the resolution. Moreover, conventional imaging suppresses high spatial frequency components close to the cutoff, making them invisible under noise-particularly the shot noise arising from discrete and random nature of...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

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Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

👤 Tsubasa Mitsutake 📅 2026-11-09 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

👤 Paul M Briley 📅 2026-11-10 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

Hamed Ekhtiari 2026-11-10 1 min 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

Shouping Zhao 2026-11-11 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

👤 Lei Huang 📅 2026-11-11 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

👤 Zahra Hadi Alqallaf 📅 2026-11-11 📖 1 min 📝 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

Yajing Meng 2026-11-05 1 min 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Bayesian Adaptive Stimulus Optimization in P300 Brain Computer Interfaces

Boyla O Mainsah 2026-11-10 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: For computational simplicity in implementation, the optimal stimulus can generally be approximated by finding the stimulus that has a combined probability mass closest to 0.5.

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